Multiple agent — Not yet.

Fast and Scalable Game-Theoretic Trajectory Planning With Intentional Uncertainties

Zhenmin Huang, Yusen Xie, Benshan Ma, Shaojie Shen, and Jun Ma, *Senior Member, IEEE*

***Abstract*—Trajectory planning involving multiagent interactions has been a long-standing challenge in the field of robotics, primarily burdened by the inherent yet intricate interactions among agents. While game-theoretic methods are widely acknowledged for their effectiveness in managing multiagent interactions, significant impediments persist when it comes to accommodating the intentional uncertainties of agents. In the context of intentional uncertainties, the heavy computational burdens associated with existing game-theoretic methods are induced, leading to inefficiencies and poor scalability. In this article, we propose a novel game-theoretic interactive trajectory planning method to effectively address the intentional uncertainties of agents, and it demonstrates both high efficiency and enhanced scalability. As the underpinning basis, we model the interactions between agents under intentional uncertainties as a static Bayesian game, and we show that its agent-form equivalence can be represented as a potential game under certain assumptions. The existence and attainability of the optimal interactive trajectories are illustrated, as the corresponding static Bayesian Nash equilibrium can be attained by optimizing a unified optimization problem. In addition, we present a distributed algorithm based on the dual consensus alternating direction method of multipliers tailored to the parallel solving of the problem, thereby significantly improving the scalability. The attendant outcomes from simulations and experiments demonstrate that the proposed method is effective across a range of scenarios characterized by general forms of intentional uncertainties. Its scalability surpasses that of existing centralized and decentralized baselines, allowing for real-time interactive trajectory planning in uncertain game settings.**

***Index Terms*—Alternating direction method of multipliers (ADMM), Bayesian game, game theory, multiagent system, nonconvex optimization, potential game.**

Received 29 November 2025; revised 4 March 2026; accepted 6 May 2026. Date of publication 26 May 2026; date of current version 23 June 2026. This article was recommended for publication by Associate Editor F. Morbidi and Editor M. Schwager upon evaluation of the reviewers' comments. *(Corresponding author: Jun Ma.)*

Zhenmin Huang, Yusen Xie, and Benshan Ma are with the Robotics and Autonomous Systems Thrust, The Hong Kong University of Science and Technology (Guangzhou), Guangzhou 511453, China (e-mail: zhuangdf@connect.ust.hk; yxie827@connect.hkust-gz.edu.cn; bma224@connect.hkust-gz.edu.cn).

Shaojie Shen is with the Department of Electronic and Computer Engineering, The Hong Kong University of Science and Technology, Hong Kong SAR, China (e-mail: eeshaojie@ust.hk).

Jun Ma is with the Robotics and Autonomous Systems Thrust, The Hong Kong University of Science and Technology (Guangzhou), Guangzhou 511453, China, and also with the Cheng Kar-Shun Robotics Institute, The Hong Kong University of Science and Technology, Hong Kong SAR, China (e-mail: jun.ma@ust.hk).

Source code is available online at https://github.com/zhuangdf/Potential-Bayesian-Game-release.

Digital Object Identifier 10.1109/TRO.2026.3697183

I. Introduction

RECENT research has witnessed an increasing interest in trajectory planning problems with multiagent interaction, which emerge from a wide range of fields such as swarm robotics [1], [2], cooperative autonomous driving [3], multi-robot coordination [4], and so on. Although fruitful results are obtained with various applications, this category of problems is particularly challenging and is far from being resolved, especially when a large number of agents are involved. Depending on the application, agents may share information and cooperate, or they may operate independently without communication. The latter case further complicates the problem, as each agent needs to reason about the intentions and the behaviors of others, which are highly interactive and coupled with its own behaviors. In light of this, methods based on game theory are gaining more and more attention, due to their superiority in modeling the interactions between selfish and noncommunicative agents. In a typical game-based trajectory planning method, each agent aims to obtain an optimal trajectory that maximizes its interests while subject to some mutual safety concerns like collision avoidance constraints, reaching a Nash equilibrium (NE) such that each agent's trajectory is optimal given all other agents' trajectories fixed to their NE. By obtaining the NE, prediction of rivals' behaviors and trajectory planning are performed in a coupled manner to avoid the "frozen robot" problem [5], which often occurs in trajectory planning methods that perform prediction and planning in a decoupled fashion and recognize other agents as immutable moving objects.

Despite current advances in this field, a series of persistent and significant challenges remain unsolved in developing game-based methods for multiagent trajectory planning. In particular, it is certainly unclear from the available existing reported works on how to deal with the game effectively with the incorporation of uncertainties, which are inherent in multiagent interactions due to factors such as sensing noises, bounded rationalities, and unknown intentions of the game players. Essentially, these ubiquitous uncertainties generally have significant impacts on the robustness of game-based planners, as they may lead to unpredictable behaviors of the game players. Notwithstanding its importance, incorporating uncertainties in game methods is far from being trivial, as it can easily render the game mathematically intractable. Many existing game-based methods simply avoid the mathematical difficulties by neglecting the uncertainties, as they focus on obtaining a single equilibrium with the strong assumption that all agents will follow the trajectories

1941-0468 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.

featured by that equilibrium [6], [7], [8], [9], [10]. However, this assumption is potentially erroneous, as the discrepancy between the obtained equilibrium and the actual behaviors of agents could lead to potential risk or even catastrophic results. Meanwhile, other methods endeavor to model uncertainties and/or multimodality in multiagent interaction from different perspectives, resulting in various trajectory planning methods based on stochastic game [11], [12], maximal entropy game [13], [14], multihypothesis game [15], [16], and Bayesian game [17], [18], [19], [20]. Nonetheless, they are either grounded on very strong and specific assumptions, like the Gaussian assumption, and therefore can only address limited cases of uncertainties or are only applicable to simple scenarios containing a limited number of agents with poor scalability. It is thus paramount that appropriate efforts should be pursued to develop a scalable game-based interactive planning method with general multimodal uncertainties fully addressed, and it is equally difficult to demonstrate the attainability or even the existence of an equilibrium under these game settings.

In this article, we introduce a novel game-based interactive trajectory planning method to address the aforementioned challenges. We address the uncertainties and the multimodal nature of the trajectory planning problem with multiagent interaction by formulating the interaction between different agents as a static Bayesian game, of which the corresponding open-loop equilibrium is shown to be optimal over the mathematical expectations of agents' underlying intentions. We show that, with certain practical assumptions, the Bayesian game is essentially a potential game, and the corresponding Bayesian Nash equilibrium (BNE) can be obtained by solving a unified optimization problem. Furthermore, a novel distributed algorithm based on the dual consensus alternating direction method of multipliers (ADMM) is introduced, which decouples the optimization problem and enables a parallelizable solving scheme. Evidently, the scalability is greatly enhanced, and computational efficiency is improved to meet the real-time requirement in trajectory planning applications. The contributions of this article are listed as follows.

1) We provide the key insight that the static Bayesian game for interactive trajectory planning is essentially a potential game with practical assumptions, and therefore, it is equivalent to a unified optimization problem (instead of a series of coupled optimization problems). This merit demonstrates the existence of the open-loop equilibrium and favorably facilitates the determination of the equilibrium through off-the-shelf nonlinear optimization solvers.
2) We show that interactive contingency planning, formulated as a contingency game by the existing literature, is also a potential game under certain assumptions. Therefore, conclusions drawn on the existence and attainability of equilibrium naturally hold. These results further expand the scope of this article to the field of contingency planning and facilitate the development of a tractable and scalable algorithm for solving the interactive contingency planning problem.
3) Tailored to the optimization problem arising from the potential Bayesian game, we propose a novel distributed algorithm based on the dual consensus ADMM, which exploits the sparsity in the optimization problem to obtain a parallelizable solving scheme. The proposed method demonstrates superiority in scalability and enables efficient solving for the BNE that meets the real-time requirements.
4) Simulations and real-world experiments on various scenarios are performed to showcase the superiority of the proposed method in terms of computational efficiency and scalability, and to illustrate its effectiveness in addressing general intentional uncertainties in multiagent interactions.
5) We also release an open-source multiprocess implementation of the proposed game solver that achieves real-time performance, which facilitates future research into game-based interactive trajectory planning.

II. Related Works

A. Game Planners Without Uncertainties

With the pervasive power in modeling the interaction between multiple agents, game-based methods have been well recognized as a promising way in handling interactive trajectory planning problems, with successful applications across different fields [8], [9], [10], [21], [22], [23], [24], [25], [26]. In [21], a real-time game-theoretic planner is developed for autonomous drone racing, with a sensitivity-enhanced iterative best response algorithm to obtain an approximation of the NE. The same method has been applied to autonomous car racing with successful results [22]. In [8], [9], [10], [23], [24], [25], and [26], game-based planners are developed to generate effective autonomous driving strategies across different traffic scenarios, including lane merging, roundabouts, and intersections. These studies collectively demonstrate the effectiveness of game-based approaches in addressing trajectory planning problems with multiagent interactions. Nevertheless, they only focus on obtaining a single equilibrium, while uncertainties and the multimodal nature of game solutions are largely overlooked, which could lead to suboptimal solutions in terms of mathematical expectation and risky planning results.

B. Game Planners With Uncertainties

Efforts have been put into tackling the uncertainties in multiagent games from both static and dynamic perspectives. While handling uncertainties, static game formulation assumes constant probability for the duration of the game and solve for the open-loop equilibrium, and the dynamic game formulation performs Bayesian updates to maintain the best estimation over the uncertainties through observations, resulting in closed-loop equilibrium strategies. As noted in numerous studies with dynamic game formulation, the stochastic game [11] mainly focuses on observation uncertainties, which uses local iterative dynamic programming in Gaussian belief space to solve a

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.

game-theoretic continuous partially observable Markov decision process. Meanwhile, maximum-entropy games [13], [14] attempt to address behavioral uncertainties by adding an extra entropy loss to the objective function, maintaining the diversity of agents' strategies. However, in these methods, the Gaussian assumption is made to render the game problem mathematically tractable, which limits the application scope. Notably, a recent study [27] introduces a novel algorithmic approach based on the implicit dual control paradigm, resulting in a tractable solution for a partially observable stochastic game. Apart from the methods mentioned above, some of the other works try to address intentional uncertainties of agents by investigating multihypothesis game methods [15], [28], most with static game formulation. In [15], a multihypothesis game is introduced, which models the uncertainty about the intention of other agents by constructing multiple hypotheses about the objectives and obtains an optimal trajectory with respect to the probability distribution of these hypotheses. Meanwhile, branch model-predictive control methods [29], [30] demonstrate that multihypothesis uncertainties can be better addressed by generating contingency plans, with each plan addressing a different hypothesis. Nevertheless, these methods view other agents as immutable objects and generally fail to consider the interaction between agents. To resolve this limitation, a contingency game [28] is investigated for interactive contingency planning, which allows the robot to efficiently interact with other agents by generating strategic motion plans conditioned on multiple possible intents for other actors in the scene. Still, these multihypothesis game-based methods are far from being real time, especially when a large number of agents are involved, and the scalability is not ideal. A branch of methods [17], [18], [19], [31], [32] aims to tackle the multimodal uncertainties through static Bayesian games, in which the corresponding BNE is shown to be two-way optimal over the expectations of different behavioral modes of both agents. Nonetheless, the game settings in these methods are quite simple and scenario specific. In [20], an integrated decision-making strategy is developed based on a static Bayesian game and is shown to be generalizable to different traffic scenarios. However, the scalability is poor due to the game tree setting, which generally admits exponential complexity. Overall, poor scalability and computational efficiency prevent the application of the above game planners with uncertainties in real-world situations, especially in complicated scenarios with a large number of agents.

C. General-Purpose Game Solvers for Interactive Trajectory Planning

As readily reported in the existing literature, game-based methods are known to be computationally heavy. To alleviate the computational burden and fulfill the real-time requirements that often occur in interactive trajectory planning, efforts have been put into developing fast and general game solvers. Among different types of games, linear–quadratic games (LQ games) [33] in which agents are assumed to possess linear dynamics and quadratic objectives are known to admit analytical solutions. The fact that NE strategies of finite-horizon LQ games satisfy the coupled Riccati equation enables the development of a fast game solver based on discrete-time dynamic programming. In [34], the Douglas–Rachford splitting method is utilized to solve the LQ game and obtain an equilibrium solution with linear convergence rate. Furthermore, the iterative LQ game [6] introduces a local algorithm for a variety of differential games, in which a series of approximated LQ games are solved iteratively to obtain the local NE strategies. Meanwhile, ALGAMES [7] introduce a novel game solver, which obtains the generalized Nash equilibrium (GNE) by satisfying the first-order optimality conditions with a quasi-Newton root-finding algorithm with constraints enforced using an augmented Lagrangian formulation. It claims to exhibit real-time performance in traffic scenarios with up to four players. Nonetheless, the computational efficiency of these game solvers is far from being ideal. While these game solvers provide general solutions to game-based interactive trajectory planning, they are grounded on the basis of solving a series of coupled optimization problem, which are mathematically challenging and often result in low computational efficiency. Furthermore, they adopt a centralized solving pattern that leads to unsatisfactory scalability regarding the number of agents.

D. Potential Games

To further enhance the computational efficiency, a potential game [35], [36] has been intensively investigated. Instead of solving for a series of coupled optimization problems, the equilibrium corresponding to a potential game can be obtained by solving a unified optimization problem with a collective objective function known as the potential function. This characteristic not only demonstrates the existence of equilibrium but also greatly enhances the attainability. Methods based on potential game have been applied to the field of autonomous driving with success [37], [38], [39]. Efforts of utilizing dynamic potential game for interactive trajectory planning are featured in [40] and [41]. In potential iterative linear quadratic regulator (iLQR) [41], the potential game is combined with a centralized iLQR algorithm to achieve fast interactive planning for drones. In [42], multiagent trajectory planning is formulated as a weighted constrained potential dynamic game to release the symmetric coupling requirement in the original potential game formulation. In the distributed potential iLQR [43], the large potential game is divided into a set of smaller local potential games according to agents' vicinity to enhance scalability. However, this approach is prone to suboptimality and is unable to generate trajectories over a long horizon, since the vicinity of agents may change. In general, the introduction of potential games reduces a series of coupled game problems to a unified optimization problem that is easier to solve, thus achieving better computational efficiency. Nonetheless, the problem of scalability persists, as the scale of the unified optimization problem continues to grow as the number of agents increases.

E. ADMM-Based Game Methods

To achieve better scalability, recent studies utilize distributed optimization algorithms, such as ADMM [44], to solve games

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.

with a large number of agents. There have been some rather promising preliminary studies on the application of ADMM for interactive trajectory planning, as featured in [45], [46], and [47]. Particularly in [47], the ADMM is utilized to solve the cooperative trajectory planning problem for an excessive number of swarm robots. Nevertheless, the computation efficiency is far from satisfying the real-time requirement. The attainability of NE through the ADMM is established in [48] and [49]. These works mainly focus on theoretical analysis without experiments showing whether real-time performance is attained. Meanwhile, the scenario-game ADMM [50] proposes a new sample-based approximation to a class of stochastic general-sum pure Nash games and applies a consensus ADMM algorithm to efficiently compute a GNE of the approximated game. By far, the method introduced in [51] is the closest to the proposed method in topic, such that it combines the scenario-game ADMM with the potential game to generate ramp-merging decisions for autonomous vehicles. Nevertheless, it aims to address uncertainties resulting from sensing noise instead of behavioral and intentional uncertainties corresponding to game agents, and the game formulation is largely different from ours. Based on above discussions, efforts are still required to develop an ADMM-based game-theoretic planner for interactive trajectory planning with intentional uncertainties, such that computational efficiency and scalability are enhanced to facilitate application in complicate scenarios.

III. Preliminaries

A typical game with multiple players can be featured by a series of optimization problems, such that each player aims to minimize its own cost by selecting the best strategy, subject to the strategies of other players

$$\begin{aligned} \min_{X^i} \quad & C_i(X^i, X^{-i}) \\ \text{s.t.} \quad & X^i \in \mathcal{X}^i \end{aligned} \tag{1}$$

where $i \in \mathcal{N}$; $\mathcal{N}$ is the set of participating players. C_i, X^i, and $\mathcal{X}^i$ are the cost function, the strategy, and the space of the strategy for player i, respectively. X^{-i} denotes the concatenated strategies of all other players except i. The corresponding NE, $[X^{i*}]_{i\in\mathcal{N}}$, is a solution to this set of optimization problems, such that for all $i \in \mathcal{N}$, X^{i*} is a minimizer to (1) given $X^{-i} = X^{-i*}$. Nevertheless, it is intuitive to see that these optimization problems are interdependent, since, in each problem, the optimal solution depends on the optimal solutions of other problems. This coupled nature introduced mathematical difficulties in solving for the NE solution, as it requires solving a series of coupled optimization problems simultaneously. Meanwhile, an exact potential game is a specific type of game, where the incentives of all players are aligned with a single global function, known as the potential function. As a result, the equilibrium of the game is obtainable by seeking the optima of the potential function, thus avoiding the mathematical difficulties in solving for a series of coupled optimization problem. Formally, we introduce the definition of the exact potential game as follows.

Definition 1. (Exact potential game): Let N be the number of players, $\mathcal{X}$ be the set of action profile over the action sets $\mathcal{X}^i$ of each player, and $C_i : \mathcal{X} \to \mathbb{R}$ be the utility function for all $i \in \{1, 2, \ldots, N\}$. A game with the above action profile and payoff function is said to be an exact potential game if there exists a potential function $P : \mathcal{X} \to \mathbb{R}$ such that $\forall i, \forall X^{-i} \in \mathcal{X}^{-i}, \forall X^i, \hat{X}^i \in \mathcal{X}^i$, the following condition holds:

$$P(X^i, X^{-i}) - P(\hat{X}^i, X^{-i}) = C_i(X^i, X^{-i}) - C_i(\hat{X}^i, X^{-i}). \tag{2}$$

Informally speaking, in a potential game, any change to the cost C_i of agent i resulting from the change of its own action X^i is equal to the change of the potential function P. With this property, methods in [37] and [41] clearly reveal the fact that the global optimum of the potential function is a pure-strategic NE solution to the potential game, as any unilateral change to X^i by player i cannot reduce the potential function and henceforth the cost of X^i, C^i. Therefore, the claim holds by the definition of NE.

IV. Potential Game With Intentional Uncertainties

In this research, we develop a novel parallelizable game-theoretic planner for interactive trajectory planning with intentional uncertainties. Specifically, in this section, we first introduce the static Bayesian game formulation of the interactive trajectory planning problem, and then, we show that it is equivalent to an agent-form game, which is a normal Nash game. Grounded on the symmetric assumption on coupling cost terms, we prove that the agent-form game admits a potential game structure, which favorably enables obtaining the corresponding equilibrium through solving a unified optimization problem. In addition, we show that a contingency game is also a potential game under the same assumption. While centralized solver results in poor scalability, we introduce a parallel solving scheme for the resulted potential game in Section V. We first develop a dual consensus ADMM algorithm tailored to the sparse structure of the potential game, of which the convergence is guaranteed under convex cases. While the potential game problem is often nonconvex, we develop an iterative convexification scheme through LQ approximation, which, when combined with the dual consensus ADMM algorithm, yields a parallel solving scheme for the potential game. The resulting game-theoretic planner is shown to demonstrate capacity in addressing intentional uncertainties with better computational efficiency and scalability than existing baselines.

A. Potential Bayesian Game

In this section, we formulate the interactive trajectory planning problem with intentional uncertainties as a Bayesian game with discrete type and demonstrate its equivalence to a potential game to show the accessibility of the corresponding BNE. For a scenario with N agents involved, we denote the set of participating agents as $\mathcal{N} = \{1, 2, \ldots, N\}$. To formulate the interactions between agents as a Bayesian game, we assume that each agent $i \in \mathcal{N}$ has its own type $t_i \in \tilde{T}_i$, where $\tilde{T}_i$ denotes the set of types corresponding to agent i. Note that these types of

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.

agents can refer to intentions like {straight going, left turning} or action styles like {aggressive, conservative}, where different types of agent i will generally correspond to different underlying utility functions and, hence, different behaviors. Intuitively, each agent does not know the exact types of other agents but can obtain an estimation of the probability distribution of their types through intention prediction based on historical trajectories. We formulate the interaction between agents as a Bayesian game with priors p, supposing that p is the joint probability distribution of the types of all agents, under the following assumption.

Assumption 1: A common distribution, p, over the probabilities of types of all agents can be established and is known to all agents.

Remark 1: While Assumption 1 is adopted (often implicitly) by many existing planning methods based on multihypothesis games and Bayesian games, we make this assumption explicit to ensure the rigor of the mathematical derivation. Experiments in Section VI-D will further show that the introduced method proves effective even when this assumption is violated.

We use X^{t_i} to denote the trajectory, which is the concatenated states and control inputs corresponding to type t_i of agent i. Namely, $X^{t_i} := [(x_\tau^{t_i}, u_\tau^{t_i})]_{\tau\in\mathcal{T}}$, where $x_\tau^{t_i}$ and $u_\tau^{t_i}$ are the state vector and input vector corresponding to type t_i of agent i at the time stamp τ, and $\mathcal{T} = \{0, 1, \ldots, T\}$ is the set of all time stamps. Physical constraints such as control limits and dynamic constraints can be expressed as $X^{t_i} \in \mathcal{X}^{t_i}$, where $\mathcal{X}^{t_i}$ is the feasible set of trajectories. A strategy of agent i, σ_i, can be defined as a map from the space of its type to the space of possible actions, namely, $\sigma_i(t_i) \in \mathcal{X}^{t_i}$, and further, the overall strategy σ is defined as $\sigma(t) := [\sigma_i(t_i)]_{i\in\mathcal{N}}$. The goal of solving for a Bayesian game is to obtain the corresponding BNE strategy $\sigma^* = [\sigma_i^*]_{i\in\mathcal{N}}$. We define the expected cost of agent i with type t_i, given a strategy σ, as

$$C_i(\sigma|t_i) := \sum_{t_{-i}\in\tilde{T}_{-i}} p(t_{-i}|t_i)\hat{C}_{t_i}(\sigma(t_{-i}, t_i)) \tag{3}$$

where t_{-i} is the concatenated type vector of all other agents except i, $\tilde{T}_{-i}$ is the set of concatenated type vectors of all agents except i, $p(t_{-i}|t_i)$ is the conditional distribution of t_{-i} given t_i, and $\hat{C}_{t_i}$ is the cost function corresponding to type t_i of agent i. A BNE is thus defined as follows.

Definition 2: A strategy σ^* is a (pure-strategic) BNE if for all $i \in \mathcal{N}$, for all $t_i \in \tilde{T}_i$, and all σ_i, the following inequality holds:

$$C_i(\sigma^*|t_i) \le C_i((\sigma_i, \sigma_{-i}^*)|t_i). \tag{4}$$

In other words, a BNE strategy for agent i, σ_i^*, is optimal for all its type $t_i \in \tilde{T}_i$ in terms of expected cost, given the BNE strategy σ_{-i}^* of all other agents. The above definition clearly shows the significance of a BNE strategy σ^* for an interactive trajectory planning problem with intentional uncertainties, in the sense that for an arbitrary agent i and its arbitrary intention t_i, the strategy $\sigma_i^*(t_i)$ is optimal over the conditional distribution $p(t_{-i}|t_i)$ of the intentions of all its rivals. Informally speaking, $\sigma_i^*(t_i)$ features the best action to take for agent i with intention t_i under the prior p.

Remark 2: In a BNE, each type t_i of player i admits an independent equilibrium strategy $\sigma_i^*(t_i)$. Therefore, the number of variables involved in the game problem, and henceforth, the size of the game problem, is proportional to the total number of types. In light of this, the scalability of the game solver should be discussed against the number of types instead of the number of players.

To solve for the BNE, we first introduce the corresponding agent-form game of the original Bayesian game. In the agent-form equivalence, each type of each player in the original Bayesian game is itself an independent player, which is often referred to as a type-player. In other words, the set of players in an agent-form game is given by $\cup_{i\in\mathcal{N}}\tilde{T}_i$. For a particular type-player $t_i \in \tilde{T}_i$, its expected cost is given by

$$C_{t_i}(X) := \sum_{t_{-i}\in\tilde{T}_{-i}} p(t_{-i}|t_i)\hat{C}_{t_i}(X^t) \tag{5}$$

where X is the joint actions of all type-players, and X^t is the joint actions of all type-players included in the type vector $t = (t_i, t_{-i})$. Note that a type vector t consists of exactly one type from each player $i \in \mathcal{N}$; it is not hard to tell that X^t is the agent-form correspondence of the strategy $\sigma(t)$ in the original Bayesian game, and thus, the equivalence between (3) and (5) is obvious. We define the following NE for the corresponding agent-form game.

Definition 3: A strategy, $X^* = [[X^{t_i*}]_{t_i\in\tilde{T}_i}]_{i\in\mathcal{N}}$, is a (pure-strategic) NE corresponding to the agent-form game, if for all $i \in \mathcal{N}$, for all $t_i \in \tilde{T}_i$, and all $X^{t_i} \in \mathcal{X}^{t_i}$, the following inequality holds:

$$C_{t_i}(X^*) \le C_{t_i}(X^{t_i}, X^{-t_i*}). \tag{6}$$

In the above definition, $X^{-t_i} = [X^{t_k}]_{t_k\in\cup_{i\in\mathcal{N}}\tilde{T}_i, t_k\neq t_i}$ is the joint strategy of all type-players except for type-player t_i. Note that it is different from $X^{t_{-i}} = [X^{t_k}]_{t_k\in t_{-i}}$, which is the joint strategy of all type-players in a type vector t_{-i}. With the above definition, we introduce the following theorem.

Theorem 1 (See [52, Theorem 9.51]): A strategy σ^* is a (pure-strategic) BNE in a Bayesian game with common prior if and only if the strategy $[[\sigma^*(t_i)]_{t_i\in\tilde{T}_i}]_{i\in\mathcal{N}}$ is an NE in its corresponding agent-form game.

Therefore, to solve for the BNE in a Bayesian game, we can instead solve for the NE in its agent-form equivalence, which is a solution to the following open-loop Nash Equilibrium problems (NEPs):

Problem 1:

$$\begin{aligned} \min_{X^{t_i}} \quad & C_{t_i}(X^{t_i}, X^{-t_i}) \\ \text{s.t.} \quad & X^{t_i} \in \mathcal{X}^{t_i} \end{aligned} \tag{7}$$

which is an optimization control problem for all type-players $t_i \in \tilde{T}_i, i \in \mathcal{N}$. Particularly, for an interactive trajectory planning problem, the expected cost $C_{t_i}(X^{t_i}, X^{-t_i})$ in (5) can be further given as

$$\begin{aligned} & C_{t_i}(X^{t_i}, X^{-t_i}) \\ & = \sum_{t_{-i}\in\tilde{T}_{-i}} p(t_{-i}|t_i)[c_{t_i}(X^{t_i}) + \sum_{j\neq i} c_{t_i t_j}(X^{t_i}, X^{t_j})]. \end{aligned} \tag{8}$$

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.

$c_{t_i}(X^{t_i})$ is the ego cost term related to tracking error and magnitude of control inputs. $c_{t_i t_j}(X^{t_i}, X^{t_j})$ is the pairwised coupled cost term between agent i with type t_i and agent j with type t_j, which aims to avoid collision and ensure safety.

In general, solving Problem 1 for the corresponding NE is not trivial. Nonetheless, we will show that given the specific structure of the cost defined in (8), together with several additional assumptions, the game featured by Problem 1 can be transformed into a potential game. As a result, the corresponding NE can be obtained by solving a unified optimization problem. Before showing the equivalence of the game featured by Problem 1 to a potential game, we further introduce two assumptions.

Assumption 2: The marginal probability of each type of each player is strictly greater than zero, namely, $p(t_i) > 0$ for all $t_i \in \tilde{T}_i, i \in \mathcal{N}$.

Assumption 3: The coupled terms between two type-players are symmetric, namely, $c_{t_i t_j}(X^{t_i}, X^{t_j}) = c_{t_j t_i}(X^{t_j}, X^{t_i})$ for all $t_i \in \tilde{T}_i, t_j \in \tilde{T}_j, i \in \mathcal{N}, j \in \mathcal{N}, j \neq i$.

With Assumption 2, we show that Problem 1 is equivalent to the following open-loop NEPs, where Assumption 2 is introduced merely to ensure the mathematical rigor of the equivalence:

Problem 2:

$$\min_{X^{t_i}} \quad p(t_i) C_{t_i}(X^{t_i}, X^{-t_i})$$
$$\text{s.t.} \quad X^{t_i} \in \mathcal{X}^{t_i}. \tag{9}$$

Namely, any solution to Problem 1 is also a solution to Problem 2 and vice versa. As a result, the game featured by Problem 2 has the same set of NE as the game featured by Problem 1.

With Assumption 3, we introduce the following theorem.

Theorem 2: The game featured by Problem 2 is a potential game with the following potential function:

$$P(X) = \sum_i \sum_{t_i \in \tilde{T}_i} p(t_i) c_{t_i}(X^{t_i}) + \sum_{i,j \in \mathcal{N}, j>i} \sum_{t_i \in \tilde{T}_i, t_j \in \tilde{T}_j} p(t_i, t_j) c_{t_i t_j}(X^{t_i}, X^{t_j}). \tag{10}$$

Proof: See Appendix A. □

Theorem 2, together with other mathematical derivations, reveal that the corresponding BNE can be obtained by solving the following unified optimization problem:

Problem 3:

$$\min_X \ P(X)$$
$$\text{s.t. } X^{t_i} \in \mathcal{X}^{t_i} \ \forall i \in \mathcal{N} \ \forall t_i \in \tilde{T}_i. \tag{11}$$

Remark 3: The size of the optimization variable X in Problem 3 is proportional to the total number of type-players $\sum_{i \in \mathcal{N}} |\tilde{T}_i|$. As a result, computational time to solve Problem 3 in a centralized manner will continue to grow with an increasing number of type-players. For example, solving Problem 3 with classic differential dynamic programming (DDP)/iLQR algorithms [41] results in a time complexity of $O((\sum_{i \in \mathcal{N}} |\tilde{T}_i|)^3)$.

Remark 4: While Assumption 3 is a common assumption by existing trajectory planning methods based on a potential game, it essentially requires that each pair of type-players regards their coupled effects to the same extent and thus being relatively strong. Since constraints are imposed on the form of coupled terms between type-players, the introduced framework can only address the uncertainties within the ego cost terms. Nevertheless, we hope to clarify that this does not exclude some interesting cases of uncertainty corresponding to the coupled terms, like considering the probability that an agent unilaterally disregards collision cost. In particular, we consider the agent that minimizes the following cost:

$$C_{t_i}(X^{t_i}, X^{-t_i}) = \sum_{t_{-i} \in \tilde{T}_{-i}} p(t_{-i}|t_i)[c_{t_i}(X^{t_i}) + \epsilon_{t_i} \sum_{j \neq i} c_{t_i t_j}(X^{t_i}, X^{t_j})]. \tag{12}$$

Compared with the original cost defined in (8), an extra coefficient ϵ_{t_i} is introduced to model the social politeness, such that a large ϵ_{t_i} corresponds to a conservative agent that pays more attention to collision avoidance, while a small ϵ_{t_i} corresponds to an aggressive agent. Particularly, when ϵ_{t_i} is an unproportionally small positive number, the corresponding agent will essentially disregard the collision cost. While the introduction of ϵ_{t_i} breaks Assumption 3, it is trivial to see that this cost is equivalent to the following cost:

$$C_{t_i}(X^{t_i}, X^{-t_i}) = \sum_{t_{-i} \in \tilde{T}_{-i}} p(t_{-i}|t_i) \left[\frac{1}{\epsilon_{t_i}} c_{t_i}(X^{t_i}) + \sum_{j \neq i} c_{t_i t_j}(X^{t_i}, X^{t_j}) \right] \tag{13}$$

such that the two costs share the same set of optimal solutions. Thus, the symmetric property is restored. This fact implies that some of the uncertainties corresponding to coupling cost terms (like the social politeness, or whether an agent considers collision avoidance) can be equivalently addressed by assigning different weights to the ego cost terms, and the application scope of the introduced method is thus extended.

B. Potential Contingency Game

Contingency planning, in which an agent maintains multiple contingency plans conditioned on the outcome of an uncertain event, is gaining popularity due to its abilities in addressing uncertainties and enhancing safety. In this section, we provide the key insight that the proposed method can easily be adapted to perform contingency planning by revealing its connection with a multihypothesis contingency game [28]. Specifically, we show that the contingency game is also a potential game with an extra constraint. Therefore, the discussions over the accessibility of the corresponding equilibrium hold naturally.

Consider a set of hypotheses Θ, where each hypothesis $\theta \in \Theta$ corresponds to a possible outcome of a traffic scenario, or in other words, a possible combination of intentions taken by all participating agents. While different actions may be taken by an agent under different hypotheses, we assign a type t_i^θ corresponding to each hypothesis θ for each agent $i \in \mathcal{N}$. We formulate the contingency game in a manner similar to [28],

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.

where the optimization problem corresponding to the ego agent, given the strategies of the other agents, is defined as

$$\mathcal{S}^{\mathrm{EA}}(X^{t^{\Theta}_{-\mathrm{EA}}}) := \arg\min_{X^{t^{\Theta}_{\mathrm{EA}}}} \sum_{\theta\in\Theta} p(\theta) C_{t^{\theta}_{\mathrm{EA}}}(X^{t^{\theta}_{\mathrm{EA}}}, X^{t^{\theta}_{-\mathrm{EA}}})$$
$$\text{s.t. } X^{t^{\theta}_{\mathrm{EA}}} \in \mathcal{X}^{t^{\theta}_{\mathrm{EA}}} \ \forall\theta\in\Theta$$
$$u^{t^{\theta}_{\mathrm{EA}}}_{\tau} = u^{\mathrm{EA}}_{\tau} \ \forall \tau \le \tau_b \forall\theta\in\Theta. \quad (14)$$

In particular, EA stands for ego agent. τ_b is the branching time stamp, and $p(\theta)$ is the probability of the hypothesis θ. $X^{t^{\Theta}_{\mathrm{EA}}} := \{X^{t^{\theta}_{\mathrm{EA}}}\}_{\theta\in\Theta}$ and $X^{t^{\Theta}_{-\mathrm{EA}}} := \{X^{t^{\theta}_{-\mathrm{EA}}}\}_{\theta\in\Theta}$ are the collections of trajectories corresponding to the ego agent and other agents under all hypotheses, respectively. u^{EA}_{τ} is the common control input at τ before the branching time. By solving this optimization problem, the ego agent manages to maintain an optimal plan in response to other agents for each hypothesis, where extra constraints are imposed to merge the gap between different contingency plans before branching time stamp τ_b. Meanwhile, we use OA to represent other agent, and the optimization problem corresponding to the other agent is defined as follows:

$$\mathcal{S}^{\mathrm{OA}}_{i,\theta}(X^{t^{\theta}_{-i}}) := \arg\min_{X^{t^{\theta}_{i}}} C_{t^{\theta}_{i}}(X^{t^{\theta}_{i}}, X^{t^{\theta}_{-i}}) \text{ s.t. } X^{t^{\theta}_{i}} \in \mathcal{X}^{t^{\theta}_{i}}$$
$$\forall\theta\in\Theta \forall i \in \mathcal{N}, i \neq \mathrm{EA}. \quad (15)$$

We provide the definition of the NE corresponding to the contingency game as follows.

Definition 4: A strategy $\{X^{t^{\Theta}_{\mathrm{EA}},*}, X^{t^{\Theta}_{-\mathrm{EA}},*}\}$ is an NE of the game defined by (14) and (15) if and only if: 1) $X^{t^{\Theta}_{\mathrm{EA}},*} \in \mathcal{S}^{\mathrm{EA}}(X^{t^{\Theta}_{-\mathrm{EA}},*})$; 2) $X^{t^{\theta}_{i},*} \in \mathcal{S}^{\mathrm{OA}}_{i,\theta}(X^{t^{\theta}_{-i},*}) \forall\theta\in\Theta\forall i\in\mathcal{N}, i\neq\mathrm{EA}$.

In particular, the cost function is defined as

$$C_{t^{\theta}_{i}}\left(X^{t^{\theta}_{i}}, X^{t^{\theta}_{-i}}\right) = c_{t^{\theta}_{i}}(X^{t^{\theta}_{i}}) + \sum_{j\neq i} c_{t^{\theta}_{i}t^{\theta}_{j}}\left(X^{t^{\theta}_{i}}, X^{t^{\theta}_{j}}\right). \quad (16)$$

It is not difficult to tell that this definition is in accordance with the definition provided in (13). Together with Assumptions 2 and 3, we introduce the following theorem.

Theorem 3: The contingency game featured by (14) and (15) is a potential game with the following potential function:

$$P'(X) = \sum_{i}\sum_{\theta\in\Theta} p(\theta) c_{t^{\theta}_{i}}(X^{t^{\theta}_{i}}) + \sum_{i,j\in\mathcal{N},j>i}\sum_{\theta} p(\theta) c_{t^{\theta}_{i}t^{\theta}_{j}}(X^{t^{\theta}_{i}}, X^{t^{\theta}_{j}}). \quad (17)$$

Proof: See Appendix B. □

Theorem 3 reveals that the corresponding NE can be obtained by the following unified optimization problem:

$$\textit{Problem 4}: \min_{X} P'(X)$$
$$\text{s.t. } X^{t^{\theta}_{i}} \in \mathcal{X}^{t^{\theta}_{i}} \ \ \forall i\in\mathcal{N} \ \forall\theta\in\Theta$$
$$u^{t^{\theta}_{\mathrm{EA}}}_{\tau} = u^{\mathrm{EA}}_{\tau} \ \ \forall\tau\le\tau_b\forall\theta\in\Theta. \quad (18)$$

The solution to this optimization problem can be easily obtained by off-the-shelf nonlinear optimization solvers, rendering the corresponding NE readily available. Thus, by showing the potential nature of a contingency game, we introduce a novel scheme for achieving interactive contingency planning.

Remark 5: It is not difficult to see that Problem 4 is adapted from Problem 3 by assigning a type t^{θ}_{i} corresponding to each hypothesis θ for each agent i with correlated probability distribution $p(t^{\theta_1}_{i}, t^{\theta_2}_{j}) = p(\theta)$ if $\theta_1 = \theta_2 = \theta$ and $p(t^{\theta_1}_{i}, t^{\theta_2}_{j}) = 0$ otherwise, as well as adding additional constraints to merge the gap between trajectories corresponding to different types of the ego agent before branching time stamp τ_b. These facts reveal that a contingency game can be viewed as a Bayesian game with a correlated probability distribution over types and an extra set of contingency constraints.

V. Game-Theoretic Planner With Parallelizable Solving Scheme via Dual Consensus ADMM

Previous discussions introduce a feasible way to solve the Bayesian game by examining its nature as a potential game. As a result, the corresponding BNE can be readily obtained by solving a unified optimal control problem. Nevertheless, complexity analysis reveals that it is particularly challenging to apply the introduced methodology to applications with strong real-time requirements like trajectory planning for autonomous agents. In fact, a centralized solving scheme can be unaffordable even for a small group of agents, as each agent can typically admit multiple types, and thus, the total number of type-players will remain substantial. Meanwhile, inspections of (10) reveal that the coupled cost terms are pairwise and only exist between type-players corresponding to different players. These facts favorably facilitate the development of a decentralized solving scheme, which exploits the sparsely coupled nature of the problem to enable parallel solving, and thus, the computational efficiency can be greatly enhanced. In this section, we introduce a novel decentralized optimization scheme based on the dual consensus ADMM, which enables a fully parallelizable solution to Problem 3 and is shown to be generalizable to other optimization problems with similar sparsely coupled structures like Problem 4, achieving superiority in computational efficiency and scalability in both Problems.

A. Sparse Dual Consensus ADMM

We first develop a novel dual consensus ADMM algorithm tailored to the sparse structure of Problem 3. We introduce the following definition of the indicator function.

Definition 5: The indicator function with respect to a set $\mathcal{X}$ is defined as

$$\mathcal{I}_{\mathcal{X}}(x) = \begin{cases} 0, & \text{if } x\in\mathcal{X} \\ \infty, & \text{otherwise.} \end{cases} \quad (19)$$

With such a definition, we define the individual cost term corresponding to the type-player t_i as

$$f_{t_i}(X^{t_i}) = p(t_i) c_{t_i}(X^{t_i}) + \mathcal{I}_{\mathcal{X}^{t_i}}(X^{t_i}) \quad (20)$$

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.

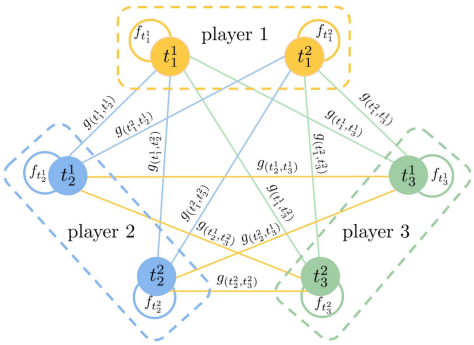


Fig. 1. Illustrative diagram of the undirected graph representing the optimization problem. Three players with altogether six types are involved, where t_1^1 and t_1^2 are the first and the second types of player 1, respectively, and other types are defined similarly. In the graph, each node represents a type-player and each edge represents a cost term involved. In particular, type-players with respect to different players are connected, while two different type-players for the same player i are not connected.

which is a combination of the weighted ego cost term and the indicator function over the feasible set. Furthermore, we assume that the coupled cost term between t_i and t_j, $c_{t_i,t_j}(X^{t_i}, X^{t_j})$, is a function over linear combination of X^{t_i} and X^{t_j}. In other words, there exists a function $g_{(t_i,t_j)}$, such that

$$g_{(t_i,t_j)}(Q_{(t_i,t_j),t_i}X^{t_i} + Q_{(t_i,t_j),t_j}X^{t_j}) = p(t_i,t_j)c_{t_i,t_j}(X^{t_i}, X^{t_j}) \quad (21)$$

for arbitrary t_i and t_j, where $Q_{(t_i,t_j),t_i}$ and $Q_{(t_i,t_j),t_j}$ are coefficient matrices. In this case, Problem 3 can be reformulated as the following optimization problem:

$$\min_{\{X^v\}} \sum_{v\in\mathcal{V}} f_v(X^v) + \sum_{e\in\mathcal{E}} g_e\left(\sum_{v\in e} Q_{v,e}X^v\right) \quad (22)$$

which can be understood as defined over an undirected graph $(\mathcal{V},\mathcal{E})$. An illustrative diagram of the undirected graph is shown in Fig. 1. In particular, $\mathcal{V}$ is the set of vertices, which in our case coincides with the set of all type-players $\cup_{i\in\mathcal{N}}\tilde{T}_i$. $\mathcal{E}$ is the set of edges such that $\mathcal{E} = \{(t_i,t_j)|t_i\in\tilde{T}_i, t_j\in\tilde{T}_j, i\in\mathcal{N}, j\in\mathcal{N}, j>i\}$, where (t_i,t_j) is an undirected edge connecting t_i and t_j. Note that only type-players with respect to different players are connected, while two different type-players t_i and t_i' for the same player i are not connected in the graph. $Q_{v,e}$ is the corresponding coefficient matrix. To separate the coupling cost terms from the individual cost terms, the problem can be reformulated as

$$\min_{\{X^v\}} \sum_{v\in\mathcal{V}} f_v(X^v) + \sum_{e\in\mathcal{E}} g_e(w_e) \text{ s.t.} \sum_{v\in e} Q_{v,e}X^v = w_e \quad (23)$$

where w_e is the independent variable corresponding to the coupled objective function g_e, whose value can be obtained by $w_e = \sum_{v\in e} Q_{v,e}X^v$. The Lagrangian is given as

$$\mathcal{L}(X,w,y) = \sum_{v\in\mathcal{V}} f_v(X^v) + \sum_{e\in\mathcal{E}} g_e(w_e) + \sum_{e\in\mathcal{E}} y_e^\top\left(\sum_{v\in e} Q_{v,e}X^v - w_e\right) \quad (24)$$

where $\{y_e\}_{e\in\mathcal{E}}$ are the Lagrange dual variables. The Lagrange dual function is thus defined as

$$\begin{aligned} h(y) &= \inf_{X,w} \mathcal{L}(X,w,y) \\ &= \inf_X \left[\sum_{v\in\mathcal{V}}(f_v(X^v) + \sum_{e\in\text{Adj}(v)} y_e^\top Q_{v,e}X^v)\right] \\ &\quad + \inf_w \left[\sum_{e\in\mathcal{E}}(g_e(w_e) - y_e^\top w_e)\right] \\ &= -\sum_{v\in\mathcal{V}} f_v^*\left(-\sum_{e\in\text{Adj}(v)} y_e^\top Q_{v,e}\right) - \sum_{e\in\mathcal{E}} g_e^*(y_e) \\ &= -\sum_{v\in\mathcal{V}}\left(f_v^*\left(-\sum_{e\in\text{Adj}(v)} y_e^\top Q_{v,e}\right) + \sum_{e\in\text{Adj}(v)} \frac{1}{N_e} g_e^*(y_e)\right) \end{aligned} \quad (25)$$

where f_v^* and g_e^* are the convex conjugate functions of f_v and g_e, respectively. N_e is the number of vertices connected by edge e. $\text{Adj}(v)$ denotes the set of edges that connect the vertex v. Maximizing the Lagrange dual function with respect to y, $\max_y h(y)$, results in the following Lagrange dual problem:

$$\min_{\{y_v\}} \sum_{v\in\mathcal{V}}(f_v^*(-Q_v^\top y_v) + g_v^*(y_v)) \text{ s.t. } y_v = [y_e]_{e\in\text{Adj}(v)}. \quad (26)$$

In particular, Q_v is the concatenated coefficient matrix defined as $Q_v = [Q_{v,e}]_{e\in\text{Adj}(v)}$. y_v is the concatenated dual variables with respect to vertex v. $g_v^*(y_v) = \sum_{e\in\text{Adj}(v)} \frac{1}{N_e} g_e^*(y_e)$ is the convex conjugate function of g_v, which is defined as $g_v(w_v) = \sum_{e\in\text{Adj}(v)} \frac{1}{N_e} g_e(N_e w_e)$. It can be seen from (26) that the Lagrange dual problem of (22) has a decomposed consensus structure. As a result, the consensus ADMM can be applied to solve this problem in a distributed manner.

To derive the corresponding consensus ADMM algorithm, we first rewrite the constraint in (26) as $E_{v,e}y_v = y_e$, where $E_{v,e}$ is a constant matrix such that by premultiplying $E_{v,e}$, the section of y_v corresponding to y_e is selected. Problem (26) can be reformulated as

$$\min_{\{y_v\}} \sum_{v\in\mathcal{V}}(\psi_v(y_v) + g_v^*(z_v)) \text{ s.t. } E_{v,e}y_v = y_e,\ y_v = z_v \quad (27)$$

where $\psi = f_v^* \circ (-Q_v^\top)$. The augmented Lagrangian corresponding to problem (26) is defined as

$$\begin{aligned} &\mathcal{L}_{\sigma,\rho}(y_v, z_v, y_e, s, \lambda) = \\ &\sum_{v\in\mathcal{V}}\left[\psi_v(y_v) + g_v^*(z_v) + s_v^\top(y_v - z_v) + \frac{\sigma}{2}||y_v - z_v||^2\right. \\ &\left. + \sum_{e\in\text{Adj}(v)} (\lambda_{v,e}^\top(E_{v,e}y_v - y_e) + \frac{\rho}{2}||E_{v,e}y_v - y_e||^2)\right] \end{aligned} \quad (28)$$

where s_v and $\lambda_{v,e}$ are dual variables. σ and ρ are positive constants. The following update steps follow from the standard

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.

ADMM algorithm

$$y_v^{k+1} = \arg\min_{y_v} \left\{ \psi_v(y_v) + s_v^{k\top} y_v + \frac{\sigma}{2}||y_v - z_v^k||^2 + \sum_{e\in \text{Adj}(v)} \left(\lambda_{v,e}^{k\top} E_{v,e} y_v + \frac{\rho}{2}||E_{v,e} y_v - y_e^k||^2 \right) \right\} \tag{29a}$$

$$z_v^{k+1} = \arg\min_{z_v} \left\{ g_v^*(z_v) - s_v^{k\top} z_v + \frac{\sigma}{2}||y_v^{k+1} - z_v||^2 \right\} \tag{29b}$$

$$y_e^{k+1} = \arg\min_{y_e} \left\{ \sum_{v\in e} -\lambda_{v,e}^{k\top} y_e + \frac{\rho}{2}||E_{v,e} y_v^{k+1} - y_e||^2 \right\} \tag{29c}$$

$$s_v^{k+1} = s_v^k + \sigma \left(y_v^{k+1} - z_v^{k+1} \right) \tag{29d}$$

$$\lambda_{v,e}^{k+1} = \lambda_{v,e}^k + \rho(E_{v,e} y_v^{k+1} - y_e^{k+1}). \tag{29e}$$

To derive a fully parallel optimization scheme with respect to vertices only, we hope to eliminate y_e and $\lambda_{v,e}$ from the above updates. The analytical solution to (29c) can be obtained as

$$y_e^{k+1} = \frac{1}{N_e} \sum_{v\in e} E_{v,e} y_v^{k+1} + \frac{1}{\rho N_e} \sum_{v\in e} \lambda_{v,e}^k. \tag{30}$$

Plugging (30) into (29e) yields

$$\lambda_{v,e}^{k+1} = \frac{N_e - 1}{N_e} \lambda_{v,e}^k - \frac{1}{N_e} \sum_{v'\in e/\{v\}} \lambda_{v',e}^k + \frac{\rho(N_e - 1)}{N_e} E_{v,e} y_v^{k+1} - \frac{\rho}{N_e} \sum_{v'\in e/\{v\}} E_{v',e} y_{v'}^{k+1} \tag{31}$$

which clearly implies that $\sum_{v\in e} \lambda_{v,e}^k = 0$ for all k. $e/\{v\}$ denotes the set of vertices connected by edge e excluding v. Therefore, (30) can be simplified as

$$y_e^{k+1} = \frac{1}{N_e} \sum_{v\in e} E_{v,e} y_v^{k+1}. \tag{32}$$

Furthermore, (29e) can be simplified as

$$\lambda_{v,e}^{k+1} = \lambda_{v,e}^k + \frac{\rho}{N_e} \sum_{v'\in e} (E_{v,e} y_v^{k+1} - E_{v',e} y_{v'}^{k+1}). \tag{33}$$

We further define $\lambda_v^k = [\lambda_{v,e}^k]_{e\in \text{Adj}(v)}$ such that $\lambda_v^{k\top} = \sum_{e\in \text{Adj}(v)} \lambda_{v,e}^{k\top} E_{v,e}$. Plugging (32) into (29a) to eliminate y_e from (29a), we have

$$y_v^{k+1} = \arg\min_{y_v} \left\{ \psi_v(y_v) + (s_v^k + \lambda_v^k)^\top y_v + \frac{\sigma}{2}||y_v - z_v^k||^2 + \sum_{e\in \text{Adj}(v)} \frac{\rho}{2}||E_{v,e} y_v - \frac{1}{N_e} \sum_{v'\in e} E_{v',e} y_{v'}^k||^2 \right\}. \tag{34}$$

Algorithm 1: Dual Consensus ADMM for Solving (22).

choose $\sigma, \rho > 0$
initialize $y_v^0, z_v^0, s_v^0, \lambda_v^0, X^{0,v}\ \forall v \in \mathcal{V}$
repeat:
  For $v \in \mathcal{V}$ **in parallel**:
    Send $E_{v,e} y_v^k$ to v' for all $e \in \text{Adj}(v), v' \in e/\{v\}$
    Update r_v^{k+1} by (36).
    Update $X^{k+1,v}$ by (38).
    Update y_v^{k+1} by (37).
    Update z_v^{k+1} by (39).
    Update s_v^{k+1} by (29d)
    For $e \in \text{Adj}(v)$:
      Update $\lambda_{v,e}^{k+1}$ by (33)
    $\lambda_v^{k+1} = [\lambda_{v,e}^{k+1}]_{e\in \text{Adj}(v)}$
until termination criterion is satisfied

By completing the square, this optimization problem is equivalent to

$$y_v^{k+1} = \arg\min_{y_v} \left\{ f_v^*(-Q_v^\top y_v) + \frac{\sigma+\rho}{2}||y_v^\top - \frac{1}{\sigma+\rho} r_v^{k+1}||^2 \right\} \tag{35}$$

where

$$r_v^{k+1} = \sigma z_v^k - \lambda_v^k - s_v^k + \sum_{e\in \text{Adj}(v)} \sum_{v'\in e} \frac{\rho}{N_e} E_{v,e}^\top E_{v',e} y_{v'}^k. \tag{36}$$

Following [53, Lemma B.1], we conclude that this optimization problem admits the solution

$$y_v^{k+1} = \frac{1}{\sigma+\rho}(Q_v X^{k+1,v} + r_v^{k+1}) \tag{37}$$

with

$$X^{k+1,v} = \arg\min_{X^v} \left\{ f_v(X^v) + \frac{1}{2(\sigma+\rho)}||Q_v X^v + r_v^{k+1}||^2 \right\}. \tag{38}$$

It will then be shown that the sequence $\{X^{k,v}\}$ will converge to a minimizer of (22). Meanwhile, by completing the square of (29b), we conclude that

$$\begin{aligned} z_v^{k+1} &= \arg\min_{y_v} \left\{ g_v^*(z_v) + \frac{\sigma}{2}||z_v - \frac{s_v^k + \sigma y_v^{k+1}}{\sigma}||^2 \right\} \\ &= \text{Prox}_{g_v^*}^{\sigma} \left(\frac{s_v^k}{\sigma} + y_v^{k+1} \right) \\ &= \frac{s_v^k}{\sigma} + y_v^{k+1} - \frac{1}{\sigma} \text{Prox}_{g_v}^{\frac{1}{\sigma}} (s_v^k + \sigma y_v^{k+1}) \end{aligned} \tag{39}$$

where the proximal operator is defined as $\text{Prox}_f^\rho(x) = \arg\min_y \{f(y) + \frac{\rho}{2}||x - y||^2\}$ and the last equation follows from [54, Theorem 14.3(ii)].

Based on the previous discussion, we introduce Algorithm 1, which is a fully distributed algorithm for solving optimization problems in the form of (22). Formally, we introduce the following assumption and theorem.

Theorem 4: Assume that f_v and g_e are convex, proper, and closed for all $v \in \mathcal{V}$ and $e \in \mathcal{E}$. Also, a primal-dual solution

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.

exists for problem (22) and strong duality holds. Furthermore, (38) admits a unique and bounded solution. The iterates $\{z_v^k\}_{v\in\mathcal{V}}$ and $\{y_v^k\}_{v\in\mathcal{V}}$ of Algorithm 1 converge to a minimizer of the dual problem (27), and the iterate $\{X^{k,v}\}_{v\in\mathcal{V}}$ of Algorithm 1 converges to a minimizer of the primal problem (22).

Proof: See Appendix C. □

For Algorithm 1, a practical termination criterion can be determined as

$$\sum_{v\in\mathcal{V}} f_v(X^{k,v}) + \sum_{e\in\mathcal{E}} g_e\left(\sum_{v\in e} Q_{v,e}X^{k,v}\right) - \sum_{v\in\mathcal{V}} f_v(X^{k+1,v}) - \sum_{e\in\mathcal{E}} g_e\left(\sum_{v\in e} Q_{v,e}X^{k+1,v}\right) < \epsilon_{\text{tol}} \quad (40)$$

where ϵ_{tol} is a tolerance number.

Particularly, similar dual consensus ADMMs are also introduced in [53] and [55]. Nevertheless, we hope to emphasize that compared to the algorithms proposed in [53] and [55], Algorithm 1 is a novel distributed algorithm for solving problems in the shape of (22), which contains individual cost terms and coupled cost terms defined in a pairwise or groupwise manner. In Algorithm 1, each node v only computes the dual variables and the auxiliary variables with respect to the coupled cost terms in which v is involved. Therefore, it is particularly suitable for solving the potential Bayesian game defined in Problem 3, where pairwise coupled cost terms exist only between pairs of type-players (t_i, t_j) corresponding to different players i and j. It is also readily applicable to other similar optimization problems that are sparsely coupled. In the simulations, we will also show that the proposed method is superior to the decentralized solving scheme based on methods proposed in [53] and [55] in terms of computational efficiency.

B. Game-Theoretic Planner With Parallel Solving Scheme

While Algorithm 1 is only applicable to convex optimization problems and Problem 3 is often nonconvex in practice, we develop an iterative convexification strategy, combined with Algorithm 1, to enable parallel solving of Problem 3. Together with the detailed definition of Problem 3, a parallelizable game-theoretic trajectory planner capable of addressing intentional uncertainties is thus introduced. Similar to [56], the individual cost term for t_i, c_{t_i}, is defined as

$$c_{t_i}(X^{t_i}) = \sum_{\tau\in\mathcal{T}} ||x_\tau^{t_i} - x_{\tau,\text{ref}}^{t_i}||_Q^2 + ||u_\tau^{t_i}||_R^2 \quad (41)$$

where $x_{\tau,\text{ref}}^{t_i}$ is the reference state vector for type-player t_i at τ. Q and R are positive-semidefinite coefficient matrices. $||\cdot||_W^2$ is the square of weighted L_2-norm $||a||_W^2 = a^\top W a$. Meanwhile, the coupled cost terms are corresponding to collision avoidance. Similar to the case of a vehicle, we represent each agent with two identical circles aligned along the longitudinal axis, with the one near the front end denoted as f and the one near the rear end denoted as r. Penalties are imposed on distances between each pair of circles corresponding to different agents, namely

$$c_{t_it_j}(X^{t_i}, X^{t_j}) = \sum_{\tau\in\mathcal{T}} \sum_{\substack{\eta,\gamma\\ \in\{f,r\}}} (l_\tau^{t_i\eta,t_j\gamma})^2$$

$$l_\tau^{t_i\eta,t_j\gamma} = \begin{cases} \sqrt{\beta}(d_\tau^{t_i\eta,t_j\gamma} - d_{\text{safe}}), & \text{if } d_\tau^{t_i\eta,t_j\gamma} < d_{\text{safe}} \\ 0, & \text{else} \end{cases} \quad (42)$$

where $d^{t_i\eta,t_j\gamma}$ is the distance between the center of circle η of t_i and circle γ of t_j, d_{safe} is the predefined safety distance, and β is a weight parameter. We define $\mathcal{X}^{t_i}$ as the feasible set of the following nonlinear dynamic constraints:

$$x_{\tau+1}^{t_i} = f(x_\tau^{t_i}, u_\tau^{t_i}). \quad (43)$$

To solve the nonconvex potential game problem, we adopt the convexification strategy similar to [56]. Namely, we perform convexification around current nominal trajectories X^{t_i} and optimize over $\delta X^{t_i} = \{\delta x_\tau^{t_i}, \delta u_\tau^{t_i}\}_{\tau\in\mathcal{T}}$. Linearization of dynamic constraints around current nominal trajectories is performed to ensure that the feasible set is linear and thus convex. Also, Gauss–Newton approximation is applied on $c_{t_it_j}$ to ensure that g is convex and (21) is satisfied. Specifically, the ego cost is reformulated as

$$\hat{c}_{t_i}(\delta X^{t_i}) = \sum_{\tau\in\mathcal{T}} ||x_\tau^{t_i} - x_{\tau,\text{ref}}^{t_i} + \delta x_\tau^{t_i}||_Q^2 + ||u_\tau^{t_i} + \delta u_\tau^{t_i}||_R^2 \quad (44)$$

and the collision avoidance cost is approximated as

$$\hat{c}_{t_it_j}(\delta X^{t_i}, \delta X^{t_j}) = \sum_{\tau\in\mathcal{T}} \sum_{\substack{\eta,\gamma\\ \in\{f,r\}}} (Q_\tau^{t_i\eta,t_j\gamma}\delta x_\tau^{t_i} + Q_\tau^{t_j\gamma,t_i\eta}\delta x_\tau^{t_j} + l_\tau^{t_i\eta,t_j\gamma})^2 \quad (45)$$

where the coefficient matrix $Q_\tau^{t_i\eta,t_j\gamma} = \partial l_\tau^{t_i\eta,t_j\gamma}/\partial x^{t_i}$ and $Q_\tau^{t_j\gamma,t_i\eta} = \partial l_\tau^{t_i\eta,t_j\gamma}/\partial x^{t_j}$. We plug the definition of $\hat{c}_{t_it_j}$ into (21) and conclude that the coefficient matrices corresponding to t_i at time stamp τ is given as $Q_{t_i,\tau} = [[\sqrt{p(t_i,t_j)}Q_\tau^{t_i\eta,t_j\gamma}]_{\eta,\gamma\in\{f,r\}}]_{t_j\in\text{adj}(t_i)}$, where $\text{adj}(t_i)$ is the set of neighbor vertices of t_i such that $t_j \in \text{adj}(t_i)$ if and only if $(t_i, t_j) \in \mathcal{E}$. Updates of r, s, y, and λ are performed in a piecewise manner regarding time stamp τ, namely

$$r_{t_i,\tau}^{k+1} = \sigma z_{t_i,\tau}^k - \lambda_{t_i,\tau}^k - s_{t_i,\tau}^k + \sum_{e\in\text{Adj}(t_i)} \sum_{t_j\in e} \frac{\rho}{N_e} E_{t_i,e,\tau}^\top E_{t_j,e,\tau} y_{t_j,\tau}^k \quad (46a)$$

$$y_{t_i,\tau}^{k+1} = \frac{1}{\sigma+\rho}\left(Q_{t_i,\tau}\delta x_\tau^{k+1,t_i} + r_{t_i,\tau}^{k+1}\right) \quad (46b)$$

$$s_{t_i,\tau}^{k+1} = s_{t_i,\tau}^k + \sigma\left(y_{t_i,\tau}^{k+1} - z_{t_i,\tau}^{k+1}\right) \quad (46c)$$

$$\lambda_{t_i,e,\tau}^{k+1} = \lambda_{t_i,e,\tau}^k + \frac{\rho}{N_e}\sum_{t_j\in e}\left(E_{t_i,e,\tau}y_{t_i,\tau}^{k+1} - E_{t_j,e,\tau}y_{t_j,\tau}^{k+1}\right). \quad (46d)$$

Likewise, $E_{t_i,e,\tau}$ is a constant matrix such that $E_{t_i,e,\tau}y_{t_i,\tau}^k$ returns the part of $y_{t_i,\tau}^k$ that correspond to coupled constraints represented by edge e. Plugging the definition of g_v in (21) into

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.

(39) yields the following z-update scheme:

$$z_{t_i,\tau}^{k+1} = \frac{1}{4\sigma+1}(4s_{t_i,\tau}^{k} + 4\sigma y_{t_i,\tau}^{k+1} + 2l_{t_i,\tau}) \tag{47}$$

where $l_{t_i,\tau} = [[\sqrt{q(t_i,t_j)}l_\tau^{t_i\eta,t_j\gamma}]_{\eta,\gamma\in\{f,r\}}]_{t_j\in\text{adj}(t_i)}$. These piecewise variables aggregate to the original r, s, y, z, and λ through concatenation as

$$\begin{aligned} &r_{t_i}^k = [r_{t_i,\tau}^k]_{\tau\in\mathcal{T}},\ z_{t_i}^k = [z_{t_i,\tau}^k]_{\tau\in\mathcal{T}},\ s_{t_i}^k = [s_{t_i,\tau}^k]_{\tau\in\mathcal{T}} \\ &y_{t_i}^k = [y_{t_i,\tau}^k]_{\tau\in\mathcal{T}}, \lambda_{t_i,\tau}^k = [\lambda_{t_i,e,\tau}^k]_{e\in\text{Adj}(t_i)},\ \lambda_{t_i}^k = [\lambda_{t_i,\tau}^k]_{\tau\in\mathcal{T}}. \end{aligned} \tag{48}$$

Meanwhile, $\delta X^{k+1,t_i}$ is the optimizer of the optimization problem in Step 7 of Algorithm 1, namely

$$\begin{aligned} &\min_{\delta X^{t_i}} \sum_{\tau\in\mathcal{T}} ||\delta x_\tau^{t_i}||_{Q\prime}^2 + q^\top \delta x_\tau^i + ||\delta u_\tau^i||_R^2 + 2u_\tau^{t_i\top} R\delta u_\tau^i \\ &\text{s.t. } \delta X^{t_i} \in \delta\mathcal{X}^{t_i} \\ &q = 2p(t_i)(x_\tau^{t_i} - x_{\tau,\text{ref}}^{t_i})^\top Q + \frac{r_{t_i,\tau}^{k+1\top} Q_{t_i,\tau}}{\sigma+\rho} \\ &Q\prime = p(t_i)Q + \frac{Q_{t_i,\tau}^\top Q_{t_i,\tau}}{2(\sigma+\rho)}. \end{aligned} \tag{49}$$

In particular, $\delta\mathcal{X}^{t_i}$ is the feasible set of the linearized kinematics. It is clear that (49) is a standard linear–quadratic regulator problem, which is readily solvable by applying dynamic programming. On solving (49), a series of feedback control strategies $\{k_\tau^{t_i}, K_\tau^{t_i}\}_{\tau\in\mathcal{T}}$ are obtained, while update of trajectories is performed by the standard iLQR update scheme together with a linear search scheme, similar to [56, Algorithm 3], which is also included here for completeness

$$\begin{aligned} u_\tau^{t_i} &\leftarrow u_\tau^{t_i} + \alpha k_\tau^{t_i} + K_\tau^{t_i}(x_\tau^{t_i} - \hat{x}_\tau^{t_i}) \\ x_{\tau+1}^{t_i} &= f(x_\tau^{t_i}, u_\tau^{t_i}). \end{aligned} \tag{50}$$

α is the line search parameter, and $\hat{x}_\tau^{t_i}$ is the state before update. Specifically, we use a discrete exhaustive line search, such that we traverse all α in a predefined table, perform updating, and select the best α that minimizes the overall cost. More details on the line search procedure can be found in [57]. Convexification, parallel solving, and trajectory updating are performed alternatively until convergence. This process is concluded as Algorithm 2. The convergence criterion in Step 2 of Algorithm 2 is that the change in value of the potential function between two consecutive iterations is smaller than 0.1. ADMM_Max_Iter is set to be 3.

Remark 6: While Problem 3 is a nonconvex optimization problem, there is no mathematical guarantee of the convergence of Algorithm 2 to a local or global optimum. Nevertheless, simulations and experimental results demonstrate that Algorithm 2 converges to a local minimizer of Problem 3 in all cases, which corresponds to a local BNE of the original Bayesian game.

Particularly for the potential contingency game defined in Problem 4, it shares the same structure as the potential Bayesian game defined in Problem 3, with extra constraints to enforce the consistency between different contingency plans. Instead of enforcing the hard constraints, we add the following soft penalties:

$$c_{\text{contingency}}(X^{t_{\text{EA}}^\Theta}) = \sum_{\theta'\neq\theta}\sum_{\tau\le\tau_b} ||x_\tau^{t_{\text{EA}}^\theta} - x_\tau^{t_{\text{EA}}^{\theta'}}||_{Q_{\text{contingency}}}^2 \tag{51}$$

where $Q_{\text{contingency}}$ is a diagonal weight matrix that penalizes inconsistency between contingency plans. Obviously, when elements of $Q_{\text{contingency}}$ go to infinity, the soft penalties become hard constraints. Due to the similarity between Problems 3 and 4, Algorithm 2 can also be applied to solve Problem 4 by adding extra edges that connect t_{EA}^θ and $t_{\text{EA}}^{\theta'}$. The decoupling of $c_{\text{contingency}}$ is similar and, therefore, is omitted for simplicity.

To enable closed-loop simulations and experiments, we adopt a strategy similar to that in [20]. After obtaining the BNE trajectories X^*, if the ego agent possesses more than one type, the best type t_i^* can be selected as

$$t_i^* = \arg\min_{t_i\in\tilde{T}_i} C_{t_i}(X^*). \tag{52}$$

With the best type t_i^*, the first step of the trajectory $X^{t_i^**}$ is performed, which resembles the receding horizon control. The common prior over the intentions of all agents is then updated by applying the Bayesian filtering on the observations of all agents' movements. Similar to [20], the Bayesian update is performed by

$$p^{\tau+H}(t) = \frac{G(\hat{x}_{\tau+H}|x_{\tau,\tau+H}^{t*},\Sigma)p^\tau(t)}{\sum_{t'\in T} G(\hat{x}_{\tau+H}|x_{\tau,\tau+H}^{t'*},\Sigma)p^\tau(t')} \tag{53}$$

where $p^\tau(t)$ is the probability of type t, H is the replanning cycle, $\hat{x}_{\tau+H}$ is the observed joint state of all agents at $\tau+H$, and $x_{\tau,\tau+H}^{t*}$ is the predicted joint state of all agents at $\tau+H$ with type vector t, obtained from the equilibrium strategy computed at τ. G denotes the joint Gaussian distribution, and Σ is the predefined covariance matrix.

Algorithm 2: Parallel Optimization for Potential Bayesian Game.

Inputs:
Connectivity graph $(\mathcal{V},\mathcal{E})$;
Reference trajectories $\{X_{\text{ref}}^{t_i}\}_{t_i\in\mathcal{V}}$;
Probabilities $\{p(t_i)\}_{t_i\in\mathcal{V}}$ and $\{p(t_i,t_j)\}_{(t_i,t_j)\in\mathcal{E}}$
Parameters $Q, R, \beta, d_{\text{safe}}, \sigma, \rho$.
Outputs:
BNE trajectories X^*.
Procedures:
1: **initialize** all trajectories $\{X^{t_i}\}$ and $P(X)$.
2: **repeat** until $|P' - P(X)| < 0.1$:
3: $P' \leftarrow P(X)$
4: **For** t^i **in parallel**:
5: Convexify around latest trajectories by (44) and (45).
6: **For** $k <$ ADMM_Max_Iter:
7: Perform updates by (46)–(49).
8: Perform trajectory updates by (50).
9: Update potential value $P(X)$.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.

VI. Simulation Results

A. Dynamic Constraints

In our simulation, we assume that all agents possess the same dynamics as a single-track bicycle, which is commonly used for vehicle dynamic constraints. Specifically, the nonlinear dynamic constraints are characterized by the following equations:

$$\begin{cases} p_{x,\tau+1} = p_{x,\tau} + f_r(v_\tau, \delta_\tau)\cos(\theta_\tau) \\ p_{y,\tau+1} = p_{y,\tau} + f_r(v_\tau, \delta_\tau)\sin(\theta_\tau) \\ \theta_{\tau+1} = \theta_\tau + \arcsin(\frac{\tau_s v_\tau \sin(\delta_\tau)}{b}) \\ v_{\tau+1} = v_\tau + \tau_s a_\tau \end{cases} \tag{54}$$

where p_x and p_y denote the X and Y coordinates of mid-point of the rear axle of the vehicle, θ is the heading angle, v is the longitudinal velocity, δ is the front wheel steering angle, and a is the acceleration. τ_s is the time interval and b is the wheelbase. f_r is defined as

$$f_r(v, \delta) = b + \tau_s v\cos(\delta) - \sqrt{b^2 - (\tau_s v \sin(\delta))^2}. \tag{55}$$

Naturally, the state vector and the control input vector are defined as $x_\tau^{t_i} = [p_{x,\tau}^{t_i}, p_{y,\tau}^{t_i}, \theta_\tau^{t_i}, v_\tau^{t_i}]$ and $u_\tau^{t_i} = [\theta_\tau^{t_i}, a_\tau^{t_i}]$, respectively.

B. Open-Loop Simulations of Potential Bayesian Games

In this section, we examine the potential Bayesian game in Problem 3 by performing open-loop simulations in different traffic scenarios (including a merging scenario and an intersection scenario). Simulation results verify the effectiveness of the introduced method in obtaining the BNE solutions across different scenarios, resulting in the optimal strategy conditioned on prior belief of the intentions of agents. The introduced method is also compared against other centralized and decentralized solving methods in terms of solution time to demonstrate its superiority in both computational efficiency and scalability. Specifically, in each scenario, we assume that the ego agent admits a unique type, while it holds a belief over the underlying types of all other agents. Through open-loop simulations, we show that a spectrum of sets of interactive trajectories can be generated by varying the belief, such that each set of interactive trajectories is a BNE corresponding to the Bayesian game under that belief. These results demonstrate the ability of the proposed method to generate optimal trajectories conditioned on the probabilities of the underlying intentions of the participating agents.

1) Qualitative Results for a Merging Scenario: We first consider a merging scenario with two agents, where the ego agent and the other agent are referred to as EA and OA, respectively. The OA intends to merge into the lane kept by the EA. The initial states of EA and OA are [0,0,0,3] and [0,4,0,3], respectively. The target of both agents is to follow the lane with $p_y = 0$. The weight matrices for both agents are defined as $Q = \text{diag}(0, 1, 0, 2)$ and $R = \text{diag}(10, 0.1)$. For collision avoidance, we set $d_{\text{safe}} = 4.5$ m and $\beta = 1.4$. We set the reference velocity of EA as 3 m/s and assume uncertainties over the reference longitudinal velocity of OA, which is subject to the following bimodal Gaussian mixture:

$$\begin{aligned} p(v_{\text{ref}}) &= w_1\mathcal{N}(v_{\text{ref}}|v_{\text{ref},1}, \sigma_1^2) + w_2\mathcal{N}(v_{\text{ref}}|v_{\text{ref},2}, \sigma_2^2) \\ v_{\text{ref},1} &= 3.5 \text{ m/s},\ v_{\text{ref},2} = 2.5 \text{ m/s}, \sigma_1 = \sigma_2 = 0.2 \text{ m/s}. \end{aligned} \tag{56}$$

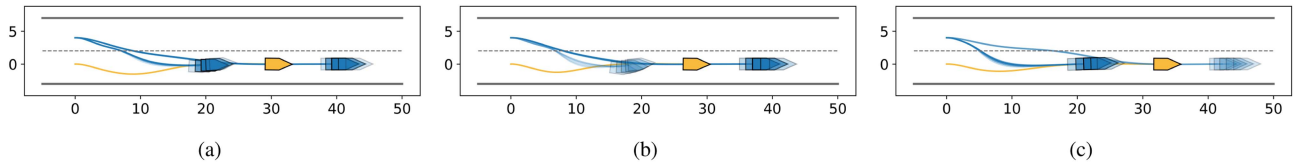
$\mathcal{N}$ denotes the Gaussian distribution, and w_1 and w_2 are weight parameters. Note that the proposed method does not preclude other forms of distribution. To obtain the optimal strategy in mathematical expectation under this uncertainty, we sample ten samples from the Gaussian mixture as $\{v_{\text{ref},i} - 2\sigma_i, v_{\text{ref},i} - \sigma_i, v_{\text{ref},i}, v_{\text{ref},i} + \sigma_i, v_{\text{ref},i} + 2\sigma_i\}$ with $i \in \{1, 2\}$ to generate the types of OA. By varying combinations of w_1 and w_2, different BNEs are obtained. The simulation results are shown in Fig. 2, where EA is yellow, and transparency is used to denote the probability of the type-players of OA. Specifically, results in Fig. 2(a) are obtained by assuming that EA holds equal beliefs over two behavioral modes of OA. In this case, the EA manages to track the preassigned reference velocity. Meanwhile, when EA has a higher belief in the higher intended velocities of OA and therefore it is more likely to merge in the front, the best strategy is to slow down to yield, resulting in slightly slower longitudinal velocities of EA compared to results of equal beliefs [see Fig. 2(b)]. In contrast, if EA believes that OA is more likely to slow down, the best strategy is to accelerate to allow OA to merge from behind, resulting in higher longitudinal velocities [see Fig. 2(c)]. These simulation results indicate that our method manages to obtain different optimal strategies conditioned on different beliefs.

2) Qualitative Results for an Intersection Scenario: We also examine an intersection scenario with three agents crossing simultaneously, where the ego agent is referred to as EA and two other agents are referred to as OA1 and OA2. The initial states of EA, OA1, and OA2 are [3, −18, 1.57, 3], [−12, −3, 0, 3], and [24, 3, 3.14, 3], respectively. EA intends to cross the intersection from the bottom to the top. OA1 intends to cross the intersection from left to right, and OA2 intends to cross the intersection from right to left. Weight matrices are given as $Q = \text{diag}(1, 1, 0, 0)$ and $R = \text{diag}(10, 0.1)$. For collision avoidance, we set $d_{\text{safe}} = 6.5$ m and $\beta = 1.4$. The reference velocity of EA is given as 3 m/s, and we assume uncertainties over the reference longitudinal velocities of both OA1 and OA2 with the following bimodal Gaussian mixture:

$$\begin{aligned} p(v_{\text{ref}}^{\text{OA1}}) &= w_1^{\text{OA1}}\mathcal{N}(v_{\text{ref}}^{\text{OA1}}|v_{\text{ref},1}^{\text{OA1}}, \sigma_{\text{OA1},1}^2) \\ &\quad + w_2^{\text{OA1}}\mathcal{N}(v_{\text{ref}}^{\text{OA1}}|v_{\text{ref},2}^{\text{OA1}}, \sigma_{\text{OA1},2}^2) \\ p(v_{\text{ref}}^{\text{OA2}}) &= w_1^{\text{OA2}}\mathcal{N}(v_{\text{ref}}^{\text{OA2}}|v_{\text{ref},1}^{\text{OA2}}, \sigma_{\text{OA2},1}^2) \\ &\quad + w_2^{\text{OA2}}\mathcal{N}(v_{\text{ref}}^{\text{OA2}}|v_{\text{ref},2}^{\text{OA2}}, \sigma_{\text{OA2},2}^2) \\ v_{\text{ref},1}^{\text{OA1}} &= v_{\text{ref},1}^{\text{OA2}} = 3.6 \text{ m/s},\ v_{\text{ref},2}^{\text{OA1}} = v_{\text{ref},2}^{\text{OA2}} = 2.4 \text{ m/s} \\ \sigma_{\text{OA1},1} &= \sigma_{\text{OA1},2} = \sigma_{\text{OA2},1} = \sigma_{\text{OA2},2} = 0.2 \text{ m/s}. \end{aligned} \tag{57}$$

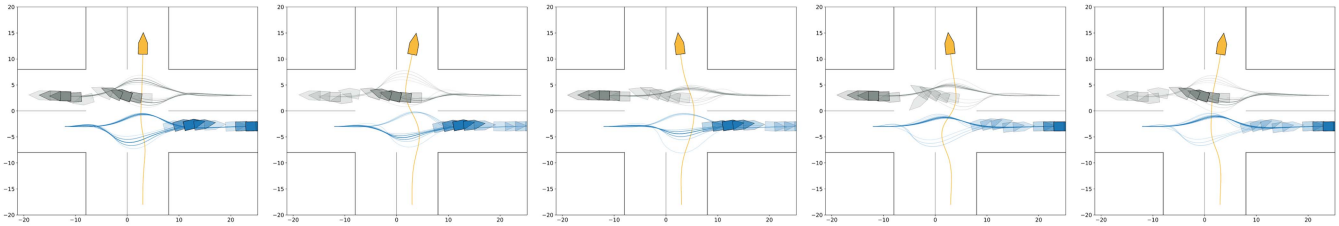
We also assume that EA admits a unique type and sample ten samples from each Gaussian mixture to obtain the types for each OA. For the EA and each type of OAs, we generate the reference trajectory by assuming uniform motion with the corresponding reference velocity. Simulation results are shown in Fig. 3. In particular, results shown in Fig. 3(a) are obtained such that equal probabilities are assumed over all behavioral modes of OAs. Compared with results of equal probabilities, results in

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c)

Fig. 2. Simulation results in the merging scenario with (a) $[w_1, w_2] = [0.5, 0.5]$, (b) $[w_1, w_2] = [0.9, 0.1]$, and (c) $[w_1, w_2] = [0.1, 0.9]$. In (a), the average longitudinal velocity of EA is roughly equal to 3 m/s, which is the reference velocity. In (b), the average longitudinal velocity of EA is smaller than 3 m/s, indicating that it slows down to yield to OA. In (c), the average longitudinal velocity of EA is higher than 3 m/s, indicating that it speeds up for OA to merge from behind.



(a) (b) (c) (d) (e)

Fig. 3. Simulation results in the intersection scenario with (a) $[w_1^{\mathrm{OA1}}, w_2^{\mathrm{OA1}}, w_1^{\mathrm{OA2}}, w_2^{\mathrm{OA2}}] = [0.5, 0.5, 0.5, 0.5]$, (b) $[w_1^{\mathrm{OA1}}, w_2^{\mathrm{OA1}}, w_1^{\mathrm{OA2}}, w_2^{\mathrm{OA2}}] = [0.1, 0.9, 0.1, 0.9]$, (c) $[w_1^{\mathrm{OA1}}, w_2^{\mathrm{OA1}}, w_1^{\mathrm{OA2}}, w_2^{\mathrm{OA2}}] = [0.1, 0.9, 0.9, 0.1]$, (d) $[w_1^{\mathrm{OA1}}, w_2^{\mathrm{OA1}}, w_1^{\mathrm{OA2}}, w_2^{\mathrm{OA2}}] = [0.9, 0.1, 0.9, 0.1]$, and (e) $[w_1^{\mathrm{OA1}}, w_2^{\mathrm{OA1}}, w_1^{\mathrm{OA2}}, w_2^{\mathrm{OA2}}] = [0.9, 0.1, 0.1, 0.9]$. In (a), equal probabilities are assumed for all behavioral modes of OAs. EA adopts an almost straight trajectory to cross the intersection scenario. In (b), both OAs are likely to slow down and yield, so EA first swerves to the right to cross in front of OA1 and then swerves to the left to cross in front of OA2. In (c), OA1 is likely to yield, and OA2 is likely to rush, so EA swerves to the right to cross in front of OA1 and from behind OA2. In (d), both OAs are likely to rush, so EA first swerves to the left to cross from behind OA1 and then swerves to the right to cross from behind OA2. In (e), OA1 is likely to rush, and OA2 is likely to yield, so EA swerves to the left to cross from behind OA1 and in front of OA2.

Fig. 3(b)–(e) show that EA decides either to rush and cross in front of the OA or to yield and cross from behind, conditioned on what it believes is the most likely behavioral mode of the OA, resulting in different optimal trajectories. These results verify the effectiveness of the proposed method in the intersection scenario by showing different sets of BNE trajectories conditioned on beliefs of the driving intentions of the OAs.

3) Computational Efficiency: Meanwhile, to demonstrate the computational efficiency and solution accuracy of the proposed method, we perform quantitative comparisons between the proposed method and the following baseline methods.

1) *Baseline 1—ILQGames:* The iLQGames [6] is a general-purpose game solver for various types of game settings. We utilize this game solver to obtain the NE corresponding to the agent-form game featured by Problem 1. We examine both Julia [iLQGames (J)] and Python [iLQGames (P)] implementations of the method.
2) *Baseline 2—IPOPT:* For this baseline, we use an interior point optimizer (IPOPT) to obtain the solution for Problem 3. The IPOPT solver is implemented by CasADi, which is an open-source software tool for numerical optimization in general and optimal control.
3) *Baseline 3—P-iLQR:* For this baseline, we solve Problem 3 by potential iLQR [41], which is a centralized solving method that utilizes the iLQR algorithm to obtain multiagent trajectories.
4) *Baseline 4—D-iLQR:* We solve Problem 3 by the decentralized iLQR [56] algorithm, which is developed directly based on the method proposed in [53] and [55].

Among these baseline methods, D-iLQR adopts a parallel solving scheme, while all other baselines are nonparallelizable. For all methods, the prediction horizon is $T = 100$ with discretization $\tau_s = 0.1$ s. For the proposed method, we provide both a single-process implementation *Ours (S)* and a multiprocess implementation *Ours (M)* for comparison. Both implementations are implemented with Python. In the multiprocess implementation, the number of processes is set to be the number of type-players. All methods are running on a server with 2 $\times$ Intel(R) Xeon(R) Gold 6348 CPU at 2.60 GHz. The quantitative results are shown in Table I. Note that in the table, T. and C. denote the computation time and the total cost at termination, respectively. While iLQGames is the only method that solves the original agent-form game instead of the corresponding potential game, the results show that it takes significantly longer compared to all other methods. In the limited cases that it manages to converge and finish within meaningful computation time, the costs at termination show that the quality of the planning results obtained is poor. From these outcomes, we conclude that solving the Bayesian game introduced in this article is extremely hard without the merit of potential game formulation, even with a state-of-the-art general-purpose game solver like iLQGames. This finding supports our claim that the potential formulation of the Bayesian game greatly enhances the accessibility of the corresponding BNE. Meanwhile, among all methods that adopt the potential game formulation, the multiprocess implementation of the proposed method achieves the lowest computational time in all cases, which clearly demonstrates its superiority in computational efficiency. Termination costs show that the

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON OF COMPUTATION TIME AND SOLUTION ACCURACY BETWEEN ILQGAMES, IPOPT, P-ILQR, D-ILQR, AND THE PROPOSED METHOD

Scenarios		Intersection						Merging					
Num. Types		5	9	13	17	21	25	3	5	7	9	11	13
iLQGames (P)	T. (s)	253.870	—	—	—	—	—	137.240	517.269	—	—	—	—
	C.	35017.1	—	—	—	—	—	2551.46	1016.87	—	—	—	—
iLQGames (J)	T. (s)	84.933	—	—	—	—	—	4.671	20.634	71.104	—	—	—
	C.	93474.6	—	—	—	—	—	796.06	958.50	879.20	—	—	—
IPOPT	T. (s)	1.370	5.520	12.190	23.920	50.970	62.470	1.150	2.370	3.200	5.290	11.310	15.680
	C.	918.27	918.38	927.78	929.85	938.79	935.77	645.87	650.21	648.22	670.64	669.7	673.68
P-iLQR	T. (s)	0.247	0.999	5.157	12.288	20.758	40.450	0.099	0.325	0.451	1.111	2.538	3.809
	C.	919.52	920.66	981.57	983.22	988.13	997.63	646.99	781.25	801.58	764.33	772.62	769.74
D-iLQR	T. (s)	0.091	0.155	0.405	0.699	1.624	2.550	0.071	0.124	0.247	0.236	0.343	0.625
	C.	921.58	928.55	943.3	951.11	960.27	967.76	649.69	659.83	728.68	832.22	831.68	835.84
Ours (S)	T. (s)	0.336	0.545	0.712	1.020	1.250	1.650	0.277	0.441	0.537	0.712	0.860	0.999
	C.	919.93	920.01	929.99	932.63	937.26	939.64	646.94	650.9	649.12	650.75	651.04	655.86
Ours (M)	T. (s)	**0.055**	**0.077**	**0.102**	**0.119**	**0.143**	**0.182**	**0.067**	**0.088**	**0.156**	**0.190**	**0.208**	**0.231**
	C.	919.93	920.01	929.99	932.63	937.26	939.64	646.94	650.9	649.12	650.75	651.04	655.86

— stands for cases with computation time > 600 s or cases fail to converge. Fastest solving time is marked in **bold** and the best solution results are marked with underline.

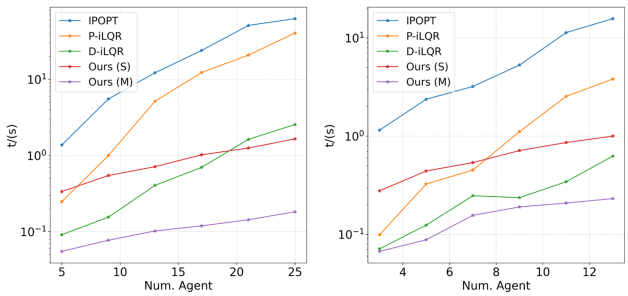


Fig. 4. Computation time with respect to the number of type-players for the baselines and the proposed method. It can be shown from the curves that the proposed method with multiprocess implementation has much better scalability than other methods in both scenarios. (a) Intersection. (b) Merging.

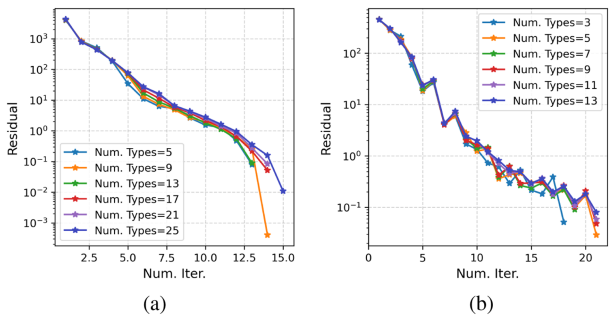


Fig. 5. Reduction in overall cost with respect to the number of iterations for the intersection and the merging scenarios. (a) Intersection. (b) Merging.

proposed method achieves optimal/near-optimal solutions in all cases. We also plot the curves of solution time with respect to the number of agents to show the scalability in Fig. 4. Quantitatively, in the intersection scenario, when the number of agent types increases by 5×, the computation time increases by 45.6×, 163.8×, 28.0×, 4.9×, and 3.3× for IPOPT, P-iLQR, D-iLQR, Ours (S), and Ours (M), respectively. In the merging scenario, when the number of agent types increases by 4.3×, the computation time increases by 13.6×, 38.5×, 8.8×, 3.6×, and 3.4×, for IPOPT, P-iLQR, D-iLQR, Ours (S), and Ours (M), respectively. These qualitative and quantitative results show the superiority of the proposed parallel solving scheme in scalability compared with all the centralized/decentralized baselines in the simulation. Meanwhile, we also plot the reduction in overall cost versus the number of iterations for both scenarios and for different numbers of types, as shown in Fig. 5. Results show that the algorithm exhibits a linear convergence rate in both scenarios. Specifically, in each scenario, the convergence rate exhibits little dependence on the number of types, which implies that the increase in computation time is mainly due to multiprocess overhead.

C. Open-Loop Simulations of Potential Contingency Game

In addition to the potential Bayesian game in Problem 3, we also examine the potential contingency game in Problem 4 to testify the capacity of the introduced method in interactive contingency planning. Specifically, we compare the introduced method with the original contingency game [28] to demonstrate that our method obtains similar results with much higher computational efficiency. We consider a two-agent overtaking scenario, in which the ego agent (referred to as EA) is trying to overtake the slower agent (referred to as OA) in front. In particular, the EA is uncertain about the target lane and target velocity of the OA, and therefore, the EA maintains a belief over the hypothesis space $\Theta = \{(p^{\text{OA}}_{y,\text{ref}}, v^{\text{OA}}_{\text{ref}}) | p^{\text{OA}}_{y,\text{ref}} \in \mathcal{R}^{\text{OA}}_{y,\text{ref}}, v^{\text{OA}}_{\text{ref}} \in \mathcal{R}^{\text{OA}}_{v,\text{ref}}\}$, where $\mathcal{R}^{\text{OA}}_{y,\text{ref}}$ and $\mathcal{R}^{\text{OA}}_{v,\text{ref}}$ are the space of possible target lanes and possible target velocities of OA, respectively. The initial states of EA and OA are given as $[-4.0, 0.5, 0, 1.0]$ and $[-2.9, 0.5, 0, 0.75]$, respectively.

We first perform a qualitative analysis to verify the effectiveness of the proposed method. We let $\mathcal{R}^{\text{OA}}_{y,\text{ref}} = \{0, 0.5\}$m and $\mathcal{R}^{\text{OA}}_{v,\text{ref}} = \{0.5\}$m/s. For each hypothesis, we set the reference lane of the EA to be different from that of the OA, namely, $p^{\text{EA}}_{y,\text{ref}} = 0$ m if $p^{\text{OA}}_{y,\text{ref}} = 0.5$ m and vice versa, so as to

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.

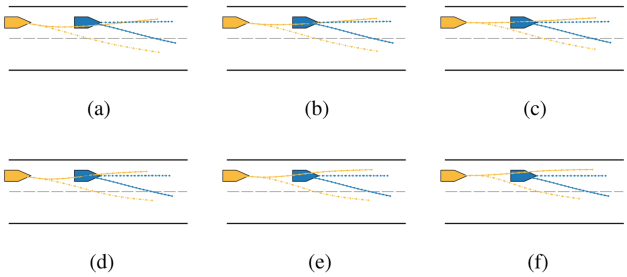


Fig. 6. Simulation results of the contingency game by the proposed method and the baseline method under different beliefs. OA is in blue, and trajectories of the same linestyle correspond to the same hypothesis. $p(\text{up})$ denotes the probability that the OA aims to follow the upper lane $p(p^{\text{OA}}_{y,\text{ref}} = 0.5\ \text{m})$. The results of the proposed method are similar to those of the baseline. When $p(\text{up})$ is high, namely that the OA is likely to follow the upper lane, the contingency plan is biased toward a lane changing decision so as to avoid the OA and perform the overtaking. As $p(\text{up})$ reduces, the contingency plan of the EA gradually shifts to favor lane keeping. These qualitative results demonstrate the effectiveness of the proposed method in interactive contingency planning. (a) Ours, $p(\text{up}) = 0.9$. (b) Ours, $p(\text{up}) = 0.5$. (c) Ours, $p(\text{up}) = 0.1$. (d) [28], $p(\text{up}) = 0.9$. (e) [28], $p(\text{up}) = 0.5$. (f) [28], $p(\text{up}) = 0.1$.

TABLE II
COMPARISON OF COMPUTATION TIME BETWEEN CONTINGENCY GAME [28] AND THE PROPOSED METHOD WITH AN INCREASING NUMBER OF HYPOTHESES

Number of hypotheses	2	4	6	8	10
Baseline [28]	0.275 s	0.286 s	0.491 s	1.199 s	22.771 s
Ours (M)	0.006 s	0.009 s	0.014 s	0.042 s	0.059 s

perform overtaking. The reference velocity of the EA is set to be 1.0 m/s for both hypotheses. We use the prediction horizon $T = 25$ and the discretization $\tau_s = 0.1$ s. Weight matrices are set to be $Q = \text{diag}(0, 0.5, 0.25, 1.0)$, $R = \text{diag}(0.5, 1.0)$, and $Q_{\text{contingency}} = \text{diag}(50, 50, 100, 10)$. For collision avoidance, we let $d_{\text{safe}} = 0.5$ m and $\beta = 1.4$. The branching time stamp is set to be $\tau_b = 5$. Results are shown in Fig. 6, where the contingency plans obtained by the proposed method and the baseline method under different beliefs are demonstrated. Similar to the baseline method, the contingency plan of the proposed method is also a function of the OA's intent probability. To facilitate overtaking, it is biased away from the lane that the OA is most likely to follow. Meanwhile, a contingency trajectory is maintained in case the anticipation of the OA's intention is wrong. These qualitative results substantiate the correctness of the potential formulation of the contingency game.

To demonstrate the efficiency and scalability of the proposed potential contingency game, we compare the solution time of the proposed method with that of the original contingency game [28] (which is nonparallelizable) with an increasing number of hypotheses. To generate up to ten hypotheses, we consider the space of possible target velocities of OA as $\mathcal{R}^{\text{OA}}_{v,\text{ref}} = \{0, 0.25, 0.5, 0.75, 1.0\}\text{m/s}$ and $\mathcal{R}^{\text{OA}}_{y,\text{ref}} = \{0, 0.5\}\text{m}$. In each case, we assume a uniform probability distribution for all hypotheses. All other parameters remain unchanged. The simulation results are shown in Table II. In our simulations, it generally takes hundreds of milliseconds to several seconds for the baseline method to converge. Meanwhile, due to the merit of the potential formulation and the parallel solving scheme introduced, the proposed method improves the computational efficiency by orders of magnitude compared with the baseline, enabling the application of the contingency game to complicated traffic scenarios with a large number of hypotheses. Also, when the number of hypotheses increases from 2 to 10, the computation time of the baseline method increases by 82.8×, while the proposed method increases by 9.8×, showing better scalability.

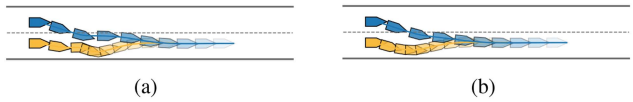


Fig. 7. Closed-loop simulation results in the merging scenario under MLE and BNE game settings with $[w_1, w_2] = [0.51, 0.49]$, and the real type of OA is that $v_{\text{ref}} = v_{\text{ref},2}$. Under the MLE game setting, the EA believes that the OA is going to slow down while in fact the OA does the opposite. Due to the misjudgment of the OA's intention, the EA first speeds up and then takes an emergency brake to avoid a collision. Meanwhile, the BNE game setting generates a trajectory that is smooth and secure for both cases. (a) MLE setting. (b) BNE setting.

D. Closed-Loop Simulations

We also perform closed-loop simulations, in which the introduced method is compared against game-based trajectory planner that only considers the most likely behavioral mode of OAs. Closed-loop simulations further demonstrate the effectiveness of the introduced method such that by properly handling intentional uncertainties, overall performances including security and passenger comforts are improved. To generate the trajectories of the OAs, we assume that they know the underlying intentions of all agents. Therefore, the OAs are playing a game without uncertainties, and they follow the NE trajectories obtained through this game. Meanwhile, the EA does not know the intentions of the OAs for certain but has access to the distribution of the underlying intentions of the OAs. Note that these settings violate Assumption 1 of common priors. We compare the performance of the EA under the following four game settings.

1) *MLE:* In this setting, the EA plays a game without uncertainties by assuming the intentions of the OAs through maximum-likelihood estimation (MLE), and follows the NE trajectory obtained through this game.
2) *BNE:* In this setting, the EA plays the Bayesian game in Problem 3 and follows the BNE trajectory.
3) *MLE-Update:* Similar to the MLE setting, the EA also plays a game without uncertainties by assuming the intentions of the OAs through MLE, but performs Bayesian filtering to update the distribution in each planning cycle. Bayesian filtering is performed based on the BNE trajectories obtained in the BNE setting.
4) *BNE-Update:* Similar to the BNE setting, the EA plays the Bayesian game and follows the BNE trajectory with Bayesian filtering to update the distribution in each planning cycle.

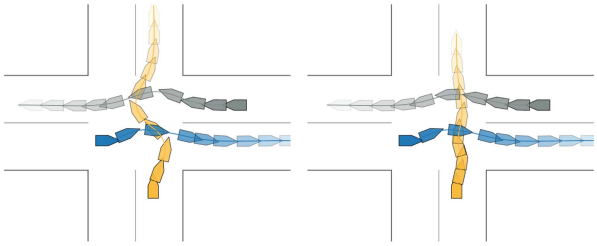
Qualitative results comparing MLE and BNE game settings are shown in Figs. 7 and 8. These results clearly demonstrate the limitation of the game method that only considers the most likely hypothesis. When there is a discrepancy between the

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.

TABLE III
COMPARISON OF PERFORMANCE BETWEEN BASELINE METHODS AND THE PROPOSED METHOD

	Intersection					Merging				
	$\|\|\tilde{\Delta V}\|\| \downarrow$ (m/s)	$\|\|\tilde{\Delta X}\|\| \downarrow$ (m)	$\|\|\tilde{\delta}\|\| \downarrow$ (rad)	$\|\|\tilde{a}\|\| \downarrow$ (m/s^2)	$\|\|\tilde{d}\|\| \uparrow$ (m)	$\|\|\tilde{\Delta V}\|\| \downarrow$ (m/s)	$\|\|\tilde{\Delta X}\|\| \downarrow$ (m)	$\|\|\tilde{\delta}\|\| \downarrow$ (rad)	$\|\|\tilde{a}\|\| \downarrow$ (m/s^2)	$\|\|\tilde{d}\|\| \uparrow$ (m)
MLE	0.568	0.810	0.202	0.991	2.389	0.486	**3.204**	0.084	0.343	2.115
BNE	0.522	**0.713**	0.196	0.866	2.828	0.458	3.307	0.074	0.291	**2.347**
MLE-Update	0.504	0.812	0.194	0.839	2.873	0.472	3.263	0.077	0.327	2.202
BNE-Update	**0.482**	0.719	**0.187**	**0.816**	**2.953**	**0.418**	3.298	**0.073**	**0.280**	2.318

Best results among the compared methods are formatted with bold and underlining.



(a) (b)

Fig. 8. Closed-loop simulation results in the intersection scenario under MLE and BNE game settings with $[w_1^{\text{OA1}}, w_2^{\text{OA1}}, w_1^{\text{OA2}}, w_2^{\text{OA2}}] = [0.51, 0.49, 0.51, 0.49]$, and the real types of OA1 and OA2 are that $v_{\text{ref}}^{\text{OA1}} = v_{\text{ref},2}^{\text{OA1}}$ and $v_{\text{ref}}^{\text{OA2}} = v_{\text{ref},2}^{\text{OA2}}$. Under the MLE game setting, the EA mistakenly believes that both OAs are going to slow down and yield. Therefore, it attempts to pass through the intersection in front of both OAs. This misjudgment leads to a collision as both OAs do the opposite. Meanwhile, the game with the BNE setting manages to generate a smooth and safe trajectory. (a) MLE setting. (b) BNE setting.

hypothesis considered by the EA and the real intention of the OAs, the trajectory generated by the game method can lead to potential danger and even collisions. On the contrary, the game method based on BNE considers different hypotheses and is well prepared for different intentions of the OAs, leading to smooth and safe trajectories. For quantitative analysis, performances of game methods under different settings are evaluated through a Monte Carlo study. For both the intersection scenario and the merging scenario, we randomly sample the initial states of both the OAs and the EA from uniform distributions centered around the initial conditions in Section VI-A. In particular, for the OAs, we also sample the distribution of their intended reference velocity by sampling the reference velocity v_{ref} and the weight parameters w_1 and w_2. For each scenario, 100 random initial conditions are drawn. Meanwhile, for each initial condition, we draw five random samples of the actual intended reference velocities of the OAs from the distribution. Altogether 500 simulations are performed for each scenario. In each simulation, we simulate the receding horizon trajectories for 10 s with the length of the planning cycle to be 2 s. In each planning cycle, the prediction horizon is $T = 100$ with discretization $\tau_s = 0.1$s. We compare the average deviation from the reference speed $||\tilde{\Delta V}||$, the average deviation from the reference trajectory $||\tilde{\Delta X}||$, the average steering angle $||\tilde{\delta}||$, and the average acceleration $||\tilde{a}||$ of the EA, as well as the average minimal distance between the EA and the OAs $||\tilde{d}||$. Simulation results are shown in Table III. Compared with the unihypothesis game solution that only considers the most likely hypothesis (the MLE method), the game solution based on BNE significantly enhances security by increasing the minimal distance between the EA and the OAs through interaction, and other performance indices are also improved. This is because the BNE trajectory of the EA is optimal in terms of mathematical expectation over the distribution of the OAs' intentions. Meanwhile, the results also verify the effectiveness of the Bayesian update in estimating the underlying intentions of the OAs; as in both the MLE and the BNE settings, the performance improves by applying the Bayesian filtering. These conclusions are further consolidated by results in Table IV, where we measure the overall cost defined by a summation of ego costs (41) and coupled costs (42) of the resulting closed-loop trajectories. These results show that the BNE solutions obtain a lower mean cost than the MLE solutions, and the solution with Bayesian updates outperforms its counterpart without Bayesian updates.

TABLE IV
COMPARISON OF BASELINE METHODS AND THE PROPOSED METHOD IN TERMS OF OVERALL COST

	MLE	BNE	MLE-Update	BNE-Update
Intersection	2521.4	2113.9	1977.7	1954.3
Merging	771.2	672.4	708.8	669.2

VII. REAL-WORLD EXPERIMENTS

We also conduct experiments to demonstrate the applicability of the introduced method under real-world situations and show that it achieves real-time performance in receding horizon planning. In the experiments, a Tianbot T100 equipped with NVIDIA Jetson Xavier NX is used as the EA, and Agilex LIMO vehicles are used as the OAs. The experiments are set up with the Robot Operating System [58] running on a laptop equipped with an 11th Gen Intel Core i7-1165G7 CPU with a clock speed of 2.80 GHz. To provide real-time feedback of the poses of all agents, we use the OptiTrack system, which operates at a frame rate of 200 frames/s. The trajectories of all agents are generated following the same scheme as in Section VI-C with the BNE game settings. Meanwhile, a pure pursuit algorithm is utilized to perform the low-level control in order to track the generated trajectories. The entire system is running at a frequency of roughly 10 Hz.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.

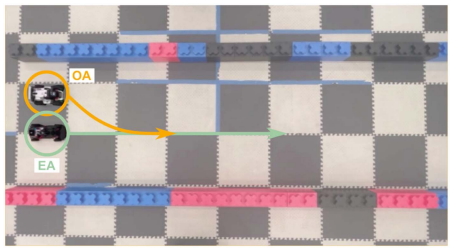


Fig. 9. Settings of the real-world experiments for the merging scenario. The EA is moving from left to right, and the OA aims to merge into the lane currently occupied by the EA. The reference speed of the EA is 0.15 m/s. The OA is either a fast-type player with a reference longitudinal speed of 0.175 m/s or a slow-type player with a reference longitudinal speed of 0.125 m/s. Equal prior probabilities are assumed for both types.

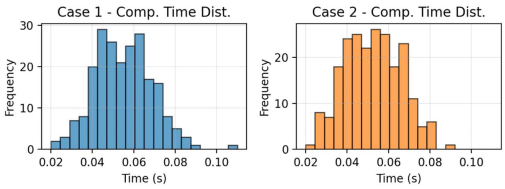
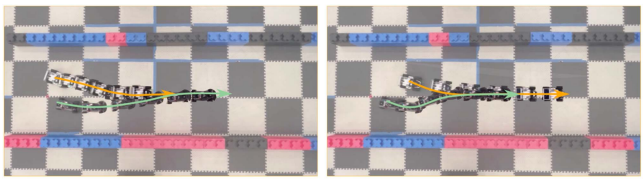


Fig. 10. Computation time distribution in real-world experiments for the merging scenario. In particular, Case 1 refers to the case where OA is a slow agent; case 2 refers to the case where OA1 is a fast agent.



(a) (b)

Fig. 11. Results of real-world experiments for the merging scenario under the case when (a) the OA is a *slow* agent and (b) the OA is a *fast* agent. In (a), the OA merges into the lower lane *from behind* the EA. In (b), the OA merges into the lower lane *in front of* the EA.

Similar to the closed-loop simulations, we also consider two scenarios, including the merging scenario and the intersection scenario. Particularly for the merging scenario, the settings of the experiments are shown in Fig. 9. In the experiments for this scenario, we examine both cases where the OA is a fast and a slow agent. We perform one trial for each case, and the total simulation time is around 10 s. Distribution of computation time for all planning cycles is plotted in Fig. 10, and the average planning time is roughly 0.05 s, which satisfies the real-time requirement. Results of experiments are shown in Fig. 11. By performing the BNE strategy, the EA properly reacts to different intentions of the OA to facilitate smooth merging. Particularly, when the OA is a fast agent, the EA slows down to yield to the OA, allowing the OA to merge in the front; instead, when the OA is a slow agent, the EA accelerates to allow the OA to merge from behind. While comparing the first snapshot of both cases, a common behavior of the EA is to steer away from the OA, which is part of the optimal maneuver obtained by computing

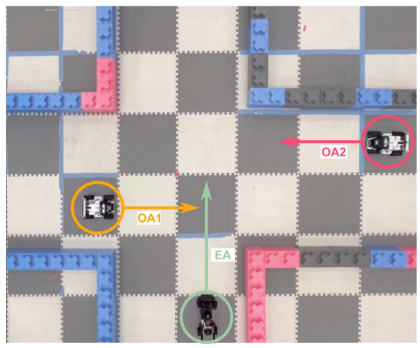


Fig. 12. Settings of the real-world experiments for the intersection scenario. The EA, OA1, and OA2 are crossing the intersection from bottom to top, from left to right, and from right to left, respectively. The reference speed of the EA is 0.15 m/s. Each OA is set to be either a fast-type player or a slow-type player. The fast-type OA aims to track a longitudinal velocity of 0.18 m/s, while the slow-type OA aims to track a longitudinal velocity of 0.12 m/s. Equal prior probabilities are assumed for all types.

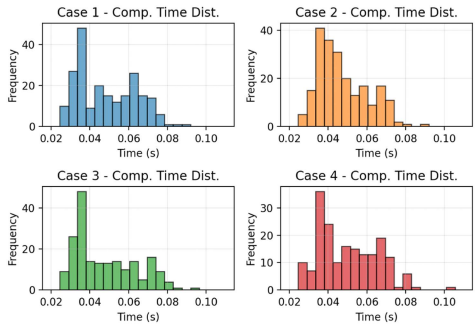


Fig. 13. Computation time distribution in real-world experiments for the intersection scenario. In particular, Case 1 refers to the case where both OAs are slow agents, Case 2 refers to the case where OA1 is slow and OA2 is fast, Case 3 refers to the case where OA1 is fast and OA2 is slow, and Case 4 refers to the case where both OAs are fast.

the BNE strategy. This enhances the safety as it increases the distance between the EA and the OA, creating enough space for the OA to merge in both cases.

For the intersection scenario, the settings are shown in Fig. 12. We enumerate all possible combinations of the types of OAs. We perform one trial for each case, and the total simulation time is around 10 s. The distribution of computation time of all planning cycles is plotted in Fig. 13. From these distributions, we conclude that the computation time of the BNE strategy is roughly 0.05 s on average, featuring real-time performance. Meanwhile, the experimental results are shown in Fig. 14. It can be seen from these trajectories that by performing the BNE strategy, the EA manages to handle different situations. Specifically, when the OA is a fast-type agent, the EA passes through the intersection from behind it; on the other hand, when the OA is a slow-type agent, the EA passes through the intersection from the front of it. A notable fact observed by comparing the first snapshot of the four different cases is that in all cases, the EA swerves to the right at the beginning, which is part of the BNE strategy given the common initial conditions and the initial beliefs over the intention of OAs in all cases. This maneuver is optimal in the sense that it steers the EA away from OA1, which is closer to the

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.

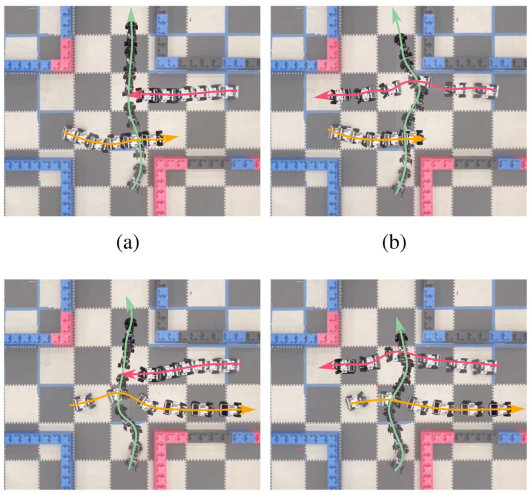


Fig. 14. Results of real-world experiments for the intersection scenario under the case when (a) both OA1 and OA2 are *slow* agents, (b) OA1 is a *slow* agent and OA2 is a *fast* agent, (c) OA1 is a *fast* agent and OA2 is a *slow* agent, and (d) both OA1 and OA2 are *fast* agents. In (a), the EA manages to pass through the intersection *in front of* both OAs. In (b), the EA passes through the intersection *in front of* OA1 and *from behind* OA2. In (c), the EA passes through the intersection *from behind* OA1 and *in front of* OA2. In (d), the EA manages to pass through the intersection *from behind* both OAs.

EA, creating space for collision avoidance and safety margins to properly react to different intentions of OAs. As a result, EA manages to pass the intersection without collisions in all cases.

To conclude, the above experiments show that the proposed method is deployable to real-world systems, and its computational efficiency meets the real-time requirement of the planning task with uncertainties in highly interactive scenarios.

VIII. Conclusion

In this work, we provide the key insight that the interactive trajectory planning problem with uncertainties, formulated as a Bayesian game, is essentially a potential game, which guarantees the existence and accessibility of the corresponding BNE by solving a unified optimization problem. With the potential game formulation, a novel optimization scheme is introduced to enable parallel solving of the game. Simulation and experimental results demonstrate the superiority of the proposed method in computational efficiency and scalability compared with existing baselines, while real-world experiments show that the proposed method is deployable to real-world systems with the real-time requirement satisfied.

While the proposed method manages to consider the intentional uncertainties of rival players, it solves a static game for open-loop equilibrium, which assumes constant probabilities of the agents' intentions through the entire game duration. Also, it does not consider other types of uncertainties that typically occur in interaction processes, such as behavioral uncertainties resulting from bounded rationality and observation uncertainties due to sensor noise. A possible future direction is to combine the proposed method with the maximum entropy model to further address fuzzy behaviors of agents. Moreover, it is also a promising direction to combine the introduced method with the partially observable Markov game, which is a dynamic game that incorporates sensor observations to update the probability of underlying variables stage by stage. As a result, a closed-loop equilibrium that performs optimal strategies with respect to sensor observations can be obtained, which may further enhance the performance of trajectory planning.

Appendix A
Proof of Theorem 2

Proof: For an arbitrary player $k \in \mathcal{N}$ and arbitrary type $t'_k \in \tilde{T}_k$, we decompose the potential function as

$$P(X) = P_{t'_k}(X^{t'_k}, X^{-t'_k}) + P_{-t'_k}(X^{-t'_k}) \tag{58}$$

where

$$\begin{aligned}
&P_{t'_k}(X^{t'_k}, X^{-t'_k}) \\
&= p(t'_k)c_{t'_k}(X^{t'_k}) + \sum_{i<k}\sum_{t_i\in\tilde{T}_i} p(t_i, t'_k)c_{t_i t'_k}(X^{t_i}, X^{t'_k}) \\
&\quad + \sum_{i>k}\sum_{t_i\in\tilde{T}_i} p(t'_k, t_i)c_{t'_k t_i}(X^{t'_k}, X^{t_i}) \\
&= p(t'_k)c_{t'_k}(X^{t'_k}) + \sum_{i\neq k}\sum_{t_i\in\tilde{T}_i} p(t'_k, t_i)c_{t'_k t_i}(X^{t'_k}, X^{t_i})
\end{aligned} \tag{59}$$

such that the second equality in (59) follows from Assumption 3. Meanwhile, for $C_{t_i}(X^{t_i}, X^{-t_i})$, we have

$$\begin{aligned}
&C_{t_i}(X^{t_i}, X^{-t_i}) \\
&= \sum_{t_{-i}\in\tilde{T}_{-i}} p(t_{-i}|t_i)[c_{t_i}(X^{t_i}) + \sum_{j\neq i} c_{t_i t_j}(X^{t_i}, X^{t_j})] \\
&= c_{t_i}(X^{t_i}) + \sum_{t_{-i}\in\tilde{T}_{-i}} p(t_{-i}|t_i)\sum_{j\neq i} c_{t_i t_j}(X^{t_i}, X^{t_j}) \\
&= c_{t_i}(X^{t_i}) + \sum_{j\neq i}\sum_{t_{-i}\in\tilde{T}_{-i}} p(t_{-i}|t_i)c_{t_i t_j}(X^{t_i}, X^{t_j}) \\
&= c_{t_i}(X^{t_i}) + \sum_{j\neq i}\sum_{t_j\in\tilde{T}_j}\sum_{t_{-ij}\in\tilde{T}_{-ij}} p(t_{-i}|t_i)c_{t_i t_j}(X^{t_i}, X^{t_j}) \\
&= c_{t_i}(X^{t_i}) + \sum_{j\neq i}\sum_{t_j\in\tilde{T}_j} p(t_j|t_i)c_{t_i t_j}(X^{t_i}, X^{t_j}).
\end{aligned} \tag{60}$$

Therefore, we have

$$P_{t'_k}(X^{t'_k}, X^{-t'_k}) = p(t'_k)C_{t'_k}(X^{t'_k}, X^{-t'_k}) \tag{61}$$

and the following conclusion:

$$\begin{aligned}
&P(X^{t'_k}, X^{-t'_k}) - P(\hat{X}^{t'_k}, X^{-t'_k}) \\
&= P_{t'_k}(X^{t'_k}, X^{-t'_k}) - P_{t'_k}(\hat{X}^{t'_k}, X^{-t'_k}) \\
&= p(t'_k)[C_{t'_k}(X^{t'_k}, X^{-t'_k}) - C_{t'_k}(\hat{X}^{t'_k}, X^{-t'_k})].
\end{aligned} \tag{62}$$

Comparing (62) and the cost function in Problem 2, we conclude that Theorem 2 follows from the fact that (62) holds for arbitrary k, t'_k, $X^{t'_k}$, $\hat{X}^{t'_k}$, and $X^{-t'_k}$. □

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.

APPENDIX B
PROOF OF THEOREM 3

Proof: We first decompose the potential function $P'(X)$ as

$$P'(X) = P'_{\text{EA}}(X^{t^{\Theta}_{\text{EA}}}, X^{t^{\Theta}_{-\text{EA}}}) + P'_{-\text{EA}}(X^{t^{\Theta}_{-\text{EA}}}) \tag{63}$$

where

$$\begin{aligned} &P'_{\text{EA}}(X^{t^{\Theta}_{\text{EA}}}, X^{t^{\Theta}_{-\text{EA}}}) \\ &= \sum_{\theta\in\Theta} p(\theta) c_{t^{\theta}_{\text{EA}}}(X^{t^{\theta}_{\text{EA}}}) + \sum_{i\neq\text{EA}} \sum_{\theta\in\Theta} p(\theta) c_{t^{\theta}_{\text{EA}} t^{\theta}_{i}}(X^{t^{\theta}_{\text{EA}}}, X^{t^{\theta}_{i}}) \\ &= \sum_{\theta\in\Theta} p(\theta) C_{t^{\theta}_{\text{EA}}}(X^{t^{\theta}_{\text{EA}}}, X^{t^{\theta}_{-\text{EA}}}) \end{aligned} \tag{64}$$

and $P'_{-\text{EA}}(X^{t^{\Theta}_{-\text{EA}}})$ is irrelevant to $X^{t^{\Theta}_{\text{EA}}}$. This implies that

$$\begin{aligned} &P'(X^{t^{\Theta}_{\text{EA}}}, X^{t^{\Theta}_{-\text{EA}}}) - P'(\hat{X}^{t^{\Theta}_{\text{EA}}}, X^{t^{\Theta}_{-\text{EA}}}) \\ &= P'_{\text{EA}}(X^{t^{\Theta}_{\text{EA}}}, X^{t^{\Theta}_{-\text{EA}}}) - P'_{\text{EA}}(\hat{X}^{t^{\Theta}_{\text{EA}}}, X^{t^{\Theta}_{-\text{EA}}}) \\ &= \sum_{\theta\in\Theta} p(\theta)[C_{t^{\theta}_{\text{EA}}}(X^{t^{\theta}_{\text{EA}}}, X^{t^{\theta}_{-\text{EA}}}) - C_{t^{\theta}_{\text{EA}}}(\hat{X}^{t^{\theta}_{\text{EA}}}, X^{t^{\theta}_{-\text{EA}}})]. \end{aligned} \tag{65}$$

Meanwhile, for $i \neq \text{EA}$ and for arbitrary $\theta \in \Theta$, we have

$$P'(X) = P'_{i,\theta}(X^{t^{\theta}_{i}}, X^{t^{\theta}_{-i}}) + P'_{-i,\theta}(X^{t^{\theta}_{-i}}) + P'_{-\theta}(X^{t^{-\theta}}) \tag{66}$$

where

$$\begin{aligned} P'_{i,\theta}(X^{t^{\theta}_{i}}, X^{t^{\theta}_{-i}}) &= p(\theta) c_{t^{\theta}_{i}}(X^{t^{\theta}_{i}}) + \sum_{j\neq i} p(\theta) c_{t^{\theta}_{i} t^{\theta}_{j}}(X^{t^{\theta}_{i}}, X^{t^{\theta}_{j}}) \\ &= p(\theta) C_{t^{\theta}_{i}}(X^{t^{\theta}_{i}}, X^{t^{\theta}_{-i}}) \end{aligned} \tag{67}$$

and the other two terms are irrelevant to $X^{t^{\theta}_{i}}$. Again, we have

$$\begin{aligned} &P'(X^{t^{\theta}_{i}}, X^{t^{\theta}_{-i}}, X^{t^{-\theta}}) - P'(\hat{X}^{t^{\theta}_{i}}, X^{t^{\theta}_{-i}}, X^{t^{-\theta}}) \\ &= P'_{i,\theta}(X^{t^{\theta}_{i}}, X^{t^{\theta}_{-i}}) - P'_{i,\theta}(\hat{X}^{t^{\theta}_{i}}, X^{t^{\theta}_{-i}}) \\ &= p(\theta)[C_{t^{\theta}_{i}}(X^{t^{\theta}_{i}}, X^{t^{\theta}_{-i}}) - C_{t^{\theta}_{i}}(\hat{X}^{t^{\theta}_{i}}, X^{t^{\theta}_{-i}})]. \end{aligned} \tag{68}$$

We also notice that fact that (15) is equivalent to the following problem with $p(\theta) > 0$:

$$\begin{aligned} \mathcal{S}^{OA}_{i,\theta}(X^{t^{\theta}_{-i}}) := &\arg\min_{X^{t^{\theta}_{i}}} p(\theta) C_{t^{\theta}_{i}}(X^{t^{\theta}_{i}}, X^{t^{\theta}_{-i}}) \text{ s.t. } X^{t^{\theta}_{i}} \in \mathcal{X}^{t^{\theta}_{i}}, \\ &\forall\theta\in\Theta,\ \forall i\in\mathcal{N}, i\neq\text{EA}. \end{aligned} \tag{69}$$

Comparing (65) and (68) with the cost of (14) and (69), respectively, we conclude that Theorem 3 holds. □

APPENDIX C
PROOF OF THEOREM 4

Proof: Algorithm 1 is derived by directly applying the ADMM algorithm to problem (27), and following [54, Cor. 28.3], it holds that the iterates $\{y^k_v\}$ and $\{z^k_v\}$ converge to y^*_v for all $v \in \mathcal{V}$, such that $[y^*_v]_{v\in\mathcal{V}}$ is a minimizer of the dual problem (26). Furthermore, the first-order optimality conditions corresponding to a primal-dual solution of problem (22) are given by

$$0 \in \partial_{X^v}\mathcal{L}(X,w,y) = \partial f_v(X^v) + \sum_{e\in\text{Adj}(v)} Q^{\top}_{v,e} y_e \quad \forall v\in\mathcal{V} \tag{70a}$$

$$0 \in \partial_{w_e}\mathcal{L}(X,w,y) = \partial g_e(w_e) - y_e \quad \forall e\in\mathcal{E} \tag{70b}$$

$$0 = \nabla_{y_e}\mathcal{L}(X,w,y) = \sum_{v\in e} Q_{v,e} X^v - w_e. \tag{70c}$$

We denote $w^k_e := \sum_{v\in e} E_{v,e} s^k_v$, $w^*_e := \lim_{k\to\infty} w^k_e$, $y^*_e := E_{v,e} y^*_v$ for arbitrary $v \in e$, and $X^{v*} := \lim_{k\to\infty} X^{k,v}$. It suffices to prove Theorem 4 by showing that (X^{v*}, w^*_e, y^*_e) satisfies (70). Step 9 of Algorithm 1 ensures that $X^{k,v}$ is a minimizer of (38) and hence

$$\begin{aligned} 0 &\in \partial f_v(X^{k,v}) + \frac{1}{\sigma+\rho} Q^{\top}_v (Q_v X^{k,v} + r^k_v) \\ &= \partial f_v(X^{k,v}) + Q^{\top}_v y^k_v \end{aligned} \tag{71}$$

where the equality follows from (37) and the conclusion holds for all k and v. Taking the limit $k \to \infty$ yields

$$0 \in \partial f_v(X^{v*}) + Q^{\top}_v y^*_v = \partial f_v(X^{v*}) + \sum_{e\in\text{Adj}(v)} Q^{\top}_{v,e} y^*_e \tag{72}$$

where the equality follows from $Q_v = [Q_{v,e}]_{e\in\text{Adj}(v)}$ and $y^*_v = [y^*_e]_{e\in\text{Adj}(v)}$, thus showing (70a). Next, Step 10 of Algorithm 1 yields that

$$0 \in \partial g^*_v(z^{k+1}_v) + \sigma(z^{k+1}_v - y^{k+1}_v) - s^k_v \tag{73}$$

holds for all v and k. For a particular $e \in \mathcal{E}$, we premultiply (73) by $E_{v,e}$, sum over $v \in e$, and take the limits $k \to \infty$, which yields

$$\begin{aligned} 0 \in &\lim_{k\to\infty} \sum_{v\in e} E_{v,e} \partial g^*_v(z^{k+1}_v) + \sigma \sum_{v\in e} E_{v,e} \left(z^{k+1}_v - y^{k+1}_v\right) \\ &- \sum_{v\in e} E_{v,e} s^k_v \\ &= \sum_{v\in e} E_{v,e} \partial g^*_v(y^*_v) - w^*_e = \sum_{v\in e} \frac{1}{N_e} \partial g^*_e(y^*_e) - w^*_e. \end{aligned} \tag{74}$$

Essentially, (74) is obtained from the fact that $y^k_v, z^k_v \to y^*_v$. From (74), we have

$$w^*_e \in \partial g^*_e(y^*_e) \Rightarrow y^*_e \in \partial g_e(w^*_e) \tag{75}$$

by invoking [54, Cor. 16.30], which shows (70b). To show (70c), we first plug (36) into (37) to obtain

$$\begin{aligned} y^{k+1}_v = \frac{1}{\sigma+\rho} \Bigg(&Q_v X^{k+1,v} + \sigma z^k_v - \lambda^k_v - s^k_v \\ &+ \sum_{e'\in\text{Adj}(v)} \sum_{v'\in e'} \frac{\rho}{N_{e'}} E^{\top}_{v,e'} E_{v',e'} y^k_{v'} \Bigg). \end{aligned} \tag{76}$$

From Steps 12 and 13 of Algorithm 1, it is not difficult to see that

$$\sum_{v\in e} E_{v,e} \lambda^k_v = \sum_{v\in e} \lambda^k_{v,e} = 0 \tag{77}$$

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.

for all k and e given zero initialization. We again premultiply (76) by $E_{v,e}$, sum over $v \in e$, and take the limits $k \to \infty$, which yields

$$N_e y_e^* = \frac{1}{\sigma+\rho}\left(\sum_{v\in e} Q_{v,e}X^{v*} + \sigma N_e y_e^* - w_e^* + \sum_{v\in e}\sum_{e'\in \mathrm{Adj}(v)}\sum_{v'\in e'} \frac{\rho}{N_{e'}} E_{v,e}E_{v,e'}^\top E_{v',e'} y_{v'}^*\right). \quad (78)$$

The last term in (78) is simplified as

$$\begin{aligned}&\sum_{v\in e}\sum_{e'\in \mathrm{Adj}(v)}\sum_{v'\in e'} \frac{\rho}{N_{e'}} E_{v,e}E_{v,e'}^\top E_{v',e'} y_{v'}^* \\ &= \sum_{v\in e}\sum_{e'\in \mathrm{Adj}(v)}\sum_{v'\in e'} \frac{\rho}{N_{e'}} E_{v,e}E_{v,e'}^\top y_{e'}^* \\ &= \sum_{v\in e}\sum_{e'\in \mathrm{Adj}(v)} \rho E_{v,e}E_{v,e'}^\top y_{e'}^* = \sum_{v\in e}\rho E_{v,e} y_v^* = \rho N_e y_e^*. \quad (79)\end{aligned}$$

Plugging this result into (78) yields (70c). We conclude the proof by establishing that (X^{v*}, w_e^*, y_e^*) satisfies (70). □

REFERENCES

[1] A. E. Turgut, H. Çelikkanat, F. Gökçe, and E. Şahin, "Self-organized flocking in mobile robot swarms," *Swarm Intell.*, vol. 2, pp. 97–120, 2008.

[2] X. Zhou et al., "Swarm of micro flying robots in the wild," *Sci. Robot.*, vol. 7, no. 66, 2022, Art. no. eabm5954.

[3] Z. Huang, W. Liu, S. Shen, and J. Ma, "A universal cooperative decision-making framework for connected autonomous vehicles with generic road topologies," *IEEE Trans. Veh. Technol.*, vol. 74, no. 4, pp. 5414–5429, Apr. 2024.

[4] P. Yu and D. V. Dimarogonas, "Distributed motion coordination for multirobot systems under LTL specifications," *IEEE Trans. Robot.*, vol. 38, no. 2, pp. 1047–1062, Apr. 2022.

[5] P. Trautman and A. Krause, "Unfreezing the robot: Navigation in dense, interacting crowds," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2010, pp. 797–803.

[6] D. Fridovich-Keil, E. Ratner, L. Peters, A. D. Dragan, and C. J. Tomlin, "Efficient iterative linear-quadratic approximations for nonlinear multi-player general-sum differential games," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2020, pp. 1475–1481.

[7] S. L. Cleac'h, M. Schwager, and Z. Manchester, "ALGAMES: A fast augmented Lagrangian solver for constrained dynamic games," *Auton. Robots*, vol. 46, no. 1, pp. 201–215, 2022.

[8] P. Hang, C. Huang, Z. Hu, Y. Xing, and C. Lv, "Decision making of connected automated vehicles at an unsignalized roundabout considering personalized driving behaviours," *IEEE Trans. Veh. Technol.*, vol. 70, no. 5, pp. 4051–4064, May 2021.

[9] P. Hang, C. Lv, C. Huang, Y. Xing, and Z. Hu, "Cooperative decision making of connected automated vehicles at multi-lane merging zone: A coalitional game approach," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 4, pp. 3829–3841, Apr. 2022.

[10] J. F. Fisac, E. Bronstein, E. Stefansson, D. Sadigh, S. S. Sastry, and A. D. Dragan, "Hierarchical game-theoretic planning for autonomous vehicles," in *Proc. Int. Conf. Robot. Automat.*, 2019, pp. 9590–9596.

[11] W. Schwarting, A. Pierson, S. Karaman, and D. Rus, "Stochastic dynamic games in belief space," *IEEE Trans. Robot.*, vol. 37, no. 6, pp. 2157–2172, Dec. 2021.

[12] H. Zhong, Y. Shimizu, and J. Chen, "Chance-constrained iterative linear-quadratic stochastic games," *IEEE Robot. Automat. Lett.*, vol. 8, no. 1, pp. 440–447, Jan. 2023.

[13] N. Mehr, M. Wang, M. Bhatt, and M. Schwager, "Maximum-entropy multi-agent dynamic games: Forward and inverse solutions," *IEEE Trans. Robot.*, vol. 39, no. 3, pp. 1801–1815, Jun. 2023.

[14] O. So, K. Stachowicz, and E. A. Theodorou, "Multimodal maximum entropy dynamic games," 2022, *arXiv:2201.12925*.

[15] F. Laine, D. Fridovich-Keil, C.-Y. Chiu, and C. Tomlin, "Multi-hypothesis interactions in game-theoretic motion planning," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2021, pp. 8016–8023.

[16] O. So, P. Drews, T. Balch, V. Dimitrov, G. Rosman, and E. A. Theodorou, "MPOGames: Efficient multimodal partially observable dynamic games," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2023, pp. 3189–3196.

[17] Z. Deng, W. Hu, Y. Yang, K. Cao, D. Cao, and A. Khajepour, "Lane change decision-making with active interactions in dense highway traffic: A Bayesian game approach," in *Proc. IEEE 25th Int. Conf. Intell. Transp. Syst.*, 2022, pp. 3290–3297.

[18] T. Zhao, W. ShangGuan, L. Chai, and Y. Cao, "A non-cooperative dynamic game-based approach for lane changing decision-making in mixed traffic diversion scenarios," in *Proc. IEEE 26th Int. Conf. Intell. Transp. Syst.*, 2023, pp. 4776–4781.

[19] Y. Zhang, P. Hang, C. Huang, and C. Lv, "Human-like interactive behavior generation for autonomous vehicles: A Bayesian game-theoretic approach with turing test," *Adv. Intell. Syst.*, vol. 4, no. 5, 2022, Art. no. 2100211.

[20] Z. Huang, T. Li, S. Shen, and J. Ma, "Integrated decision making and trajectory planning for autonomous driving under multimodal uncertainties: A Bayesian game approach," *IEEE Trans. Veh. Technol.*, 2026.

[21] R. Spica, E. Cristofalo, Z. Wang, E. Montijano, and M. Schwager, "A real-time game theoretic planner for autonomous two-player drone racing," *IEEE Trans. Robot.*, vol. 36, no. 5, pp. 1389–1403, Oct. 2020.

[22] M. Wang, Z. Wang, J. Talbot, J. C. Gerdes, and M. Schwager, "Game-theoretic planning for self-driving cars in multivehicle competitive scenarios," *IEEE Trans. Robot.*, vol. 37, no. 4, pp. 1313–1325, Aug. 2021.

[23] C. Li, T. Trinh, L. Wang, C. Liu, M. Tomizuka, and W. Zhan, "Efficient game-theoretic planning with prediction heuristic for socially-compliant autonomous driving," *IEEE Robot. Automat. Lett.*, vol. 7, no. 4, pp. 10248–10255, Oct. 2022.

[24] A. Dreves and M. Gerdts, "A generalized Nash equilibrium approach for optimal control problems of autonomous cars," *Optimal Control Appl. Methods*, vol. 39, no. 1, pp. 326–342, 2018.

[25] P. Hang, C. Huang, Z. Hu, and C. Lv, "Decision making for connected automated vehicles at urban intersections considering social and individual benefits," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 22549–22562, Nov. 2022.

[26] M. Liu, Y. Wan, F. L. Lewis, S. Nageshrao, and D. Filev, "A three-level game-theoretic decision-making framework for autonomous vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 20298–20308, Nov. 2022.

[27] H. Hu, D. Isele, S. Bae, and J. F. Fisac, "Active uncertainty reduction for safe and efficient interaction planning: A shielding-aware dual control approach," *Int. J. Robot. Res.*, vol. 43, no. 9, pp. 1382–1408, 2024.

[28] L. Peters et al., "Contingency games for multi-agent interaction," *IEEE Robot. Automat. Lett.*, vol. 9, no. 3, pp. 2208–2215, Mar. 2024.

[29] L. Zhang, C. Lin, and S. Grammatico, "Parallel branch model predictive control on GPUs," 2025, *arXiv:2506.13624*.

[30] Y. Chen, U. Rosolia, W. Ubellacker, N. Csomay-Shanklin, and A. D. Ames, "Interactive multi-modal motion planning with branch model predictive control," *IEEE Robot. Automat. Lett.*, vol. 7, no. 2, pp. 5365–5372, Apr. 2022.

[31] Y. Ali, Z. Zheng, M. M. Haque, and M. Wang, "A game theory-based approach for modelling mandatory lane-changing behaviour in a connected environment," *Transp. Res. C: Emerg. Technol.*, vol. 106, pp. 220–242, 2019.

[32] H. Yu, H. E. Tseng, and R. Langari, "A human-like game theory-based controller for automatic lane changing," *Transp. Res. C: Emerg. Technol.*, vol. 88, pp. 140–158, 2018.

[33] T. Başar and G. J. Olsder, *Dynamic Noncooperative Game Theory*. Philadelphia, PA, USA: SIAM, 1998.

[34] R. R. Baghbadorani, E. Benenati, and S. Grammatico, "A Douglas-Rachford splitting method for solving monotone variational inequalities in linear-quadratic dynamic games," 2025, *arXiv:2504.05757*.

[35] D. Monderer and L. S. Shapley, "Potential games," *Games Econ. Behav.*, vol. 14, no. 1, pp. 124–143, 1996.

[36] R. W. Rosenthal, "A class of games possessing pure-strategy Nash equilibria," *Int. J. Game Theory*, vol. 2, pp. 65–67, 1973.

[37] M. Liu, I. Kolmanovsky, H. E. Tseng, S. Huang, D. Filev, and A. Girard, "Potential game-based decision-making for autonomous driving," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 8, pp. 8014–8027, Aug. 2023.

[38] M. Liu, H. E. Tseng, D. Filev, A. Girard, and I. Kolmanovsky, "Safe and human-like autonomous driving: A predictor–corrector potential game approach," *IEEE Trans. Control Syst. Technol.*, vol. 32, no. 3, pp. 834–848, May 2024.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.

[39] F. Fabiani and S. Grammatico, "Multi-vehicle automated driving as a generalized mixed-integer potential game," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 3, pp. 1064–1073, Mar. 2020.
[40] M. Bhatt, Y. Jia, and N. Mehr, "Efficient constrained multi-agent trajectory optimization using dynamic potential games," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2023, pp. 7303–7310.
[41] T. Kavuncu, A. Yaraneri, and N. Mehr, "Potential iLQR: A potential-minimizing controller for planning multi-agent interactive trajectories," in *Proc. Robot. Sci. Syst.*, Cambridge, MA, USA: MIT Press, 2021, *arXiv:2107.04926*.
[42] M. Bhatt, Y. Jia, and N. Mehr, "Strategic decision-making in multi-agent domains: A weighted constrained potential dynamic game approach," *IEEE Trans. Robot.*, vol. 41, pp. 2749–2764, 2025.
[43] Z. Williams, J. Chen, and N. Mehr, "Distributed potential iLQR: Scalable game-theoretic trajectory planning for multi-agent interactions," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2023, pp. 1–7.
[44] S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein, "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Found. Trends Mach. Learn.*, vol. 3, no. 1, pp. 1–122, 2011.
[45] X. Zhang, Z. Cheng, J. Ma, S. Huang, F. L. Lewis, and T. H. Lee, "Semi-definite relaxation-based ADMM for cooperative planning and control of connected autonomous vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 9240–9251, Jul. 2022.
[46] F. Rey, Z. Pan, A. Hauswirth, and J. Lygeros, "Fully decentralized ADMM for coordination and collision avoidance," in *Proc. Eur. Control Conf.*, 2018, pp. 825–830.
[47] A. D. Saravanos, Y. Aoyama, H. Zhu, and E. A. Theodorou, "Distributed differential dynamic programming architectures for large-scale multiagent control," *IEEE Trans. Robot.*, vol. 39, no. 6, pp. 4387–4407, Dec. 2023.
[48] E. Börgens and C. Kanzow, "ADMM-type methods for generalized Nash equilibrium problems in Hilbert spaces," *SIAM J. Optim.*, vol. 31, no. 1, pp. 377–403, 2021.
[49] F. Salehisadaghiani and L. Pavel, "Distributed Nash equilibrium seeking via the alternating direction method of multipliers," *IFAC-PapersOnLine*, vol. 50, no. 1, pp. 6166–6171, 2017.
[50] J. Li et al., "Scenario-game ADMM: A parallelized scenario-based solver for stochastic noncooperative games," in *Proc. 62nd IEEE Conf. Decis. Control*, 2023, pp. 8093–8099.
[51] D. Jing, E. Yao, R. Chen, and M. Menéndez, "Decentralized human-like ramp merging decision-making and control based on a stochastic potential game," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 10, pp. 16724–16734, Oct. 2025.
[52] M. Maschler, S. Zamir, and E. Solan, *Game Theory*. Cambridge, U.K.: Cambridge Univ. Press, 2013.
[53] G. Banjac, F. Rey, P. Goulart, and J. Lygeros, "Decentralized resource allocation via dual consensus ADMM," in *Proc. Amer. Control Conf.*, 2019, pp. 2789–2794.
[54] H. H. Bauschke and P. L. Combettes, "Correction to: Convex analysis and monotone operator theory in Hilbert spaces," in *Convex Analysis and Monotone Operator Theory in Hilbert Spaces*. Berlin, Germany: Springer, 2019.
[55] P. D. Grontas, M. W. Fisher, and F. Dörfler, "Distributed and constrained $\mathcal{H}_2$ control design via system level synthesis and dual consensus ADMM," in *Proc. IEEE 61st Conf. Decis. Control*, 2022, pp. 301–307.
[56] Z. Huang, S. Shen, and J. Ma, "Decentralized iLQR for cooperative trajectory planning of connected autonomous vehicles via dual consensus ADMM," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 11, pp. 12754–12766, Nov. 2023.
[57] J. Nocedal and S. J. Wright, *Numerical Optimization*. Berlin, Germany: Springer, 2006.
[58] M. Quigley et al., "ROS: An open-source robot operating system," *ICRA Workshop Open Source Softw.*, vol. 3, no. 3.2, 2009, Art. no. 5.



Zhenmin Huang received the B.Eng. degree in mechanical engineering from Shanghai Jiao Tong University, Shanghai, China, in 2018, and the M.Sc. degree in mechanical engineering from Purdue University, West Lafayette, IN, USA, in 2019. He is currently working toward the Ph.D. degree in robotics and autonomous systems with The Hong Kong University of Science and Technology, Hong Kong SAR, China.

His research interests include motion planning, optimal control, optimization, multiagent systems, and autonomous driving.



Yusen Xie received the B.Sc. degree in geographic information systems from Wuhan University of Technology, Wuhan, China, in 2020, and the M.Sc. degree in electronic and information engineering from Beijing Information Science and Technology University, Beijing, China, in 2023. He is currently working toward the Ph.D. degree in robotics and autonomous systems with The Hong Kong University of Science and Technology (Guangzhou), Guangzhou, China.

His research interests include multisensor simultaneous localization and mapping, perception, and autonomous driving.



Benshan Ma received the B.Eng. degree from Chongqing University, Chongqing, China, in 2020, and the M.Sc. degree from Beijing Institute of Technology, Beijing, China, in 2023, both in vehicle engineering. He is currently working toward the Ph.D. degree in robotics and autonomous systems with The Hong Kong University of Science and Technology (Guangzhou), Guangzhou, China.

His research interests include decision making and motion planning in autonomous driving.



Shaojie Shen received the B.Eng. degree in electronic engineering from The Hong Kong University of Science and Technology (HKUST), Hong Kong, in 2009, and the M.S. degree in robotics and the Ph.D. degree in electrical and systems engineering from the University of Pennsylvania, Philadelphia, PA, USA, in 2011 and 2014, respectively.

In 2014, he joined the Department of Electronic and Computer Engineering, HKUST, as an Assistant Professor and became an Associate Professor in 2020. He is currently the Founding Director of the DJI Joint Innovation Laboratory, HKUST. His research interests include robotics and uncrewed aerial agents, especially state estimation, sensor fusion, localization and mapping, and autonomous navigation in complex environments.



Jun Ma (Senior Member, IEEE) received the B.Eng. degree with First Class Honours in electrical and electronic engineering from Nanyang Technological University, Singapore, in 2014, and the Ph.D. degree in electrical and computer engineering from the National University of Singapore, Singapore, in 2018.

From 2018 to 2021, he held several positions with the National University of Singapore, Singapore; University College London, London, U.K.; University of California, Berkeley, CA, USA; and Harvard University, Cambridge, MA, USA. He is currently an Assistant Professor with the Robotics and Autonomous Systems Thrust, The Hong Kong University of Science and Technology (Guangzhou), Guangzhou, China, and also with the Cheng Kar-Shun Robotics Institute, The Hong Kong University of Science and Technology, Hong Kong SAR, China. He is also the Director of Intelligent Autonomous Driving Center, The Hong Kong University of Science and Technology (Guangzhou). His research interests include motion planning and control for robotics and autonomous driving.

Dr. Ma was the recipient of the Honorable Mention for the 2024 IEEE Robotics and Automation Letters Best Paper Award, 2025 IEEE ICRA Best Conference Paper Award Finalist, 2025 IEEE IROS Best Student Paper Award Finalist and Best Paper Award on Mobile Manipulation, etc. He was also a recipient of the ACA Temasek Young Educator Award.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:36 UTC from IEEE Xplore. Restrictions apply.